

FERNS OF HAWAII NATIONAL PARK



Special Issue

HAWAII NATURE NOTES

Price

FERNS OF HAWAII NATIONAL PARK



Special Issue

HAWAII NATURE NOTES

Price

The Project Gutenberg eBook of Ferns of Hawaii National Park

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Ferns of Hawaii National Park

Author: Douglass H. Hubbard

Release date: July 13, 2019 [eBook #59913]

Language: English

Other information and formats: www.gutenberg.org/ebooks/59913

Credits: Produced by Stephen Hutcheson and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK FERNS OF
HAWAII NATIONAL PARK ***

COVER

Amaumau ferns in the Kilauea Section, Hawaii National Park

DEDICATION

To MR. EUGENE HORNER, Member of the Board of Directors of the HAWAII NATURAL HISTORY ASSOCIATION, whose interest in Hawaiian ferns has extended over more than a half a century, and whose enthusiasm and willing assistance in collecting and identifying ferns has been of invaluable help, this booklet is dedicated.

PRINTED BY
14ND PUBLICATIONS & PRINTING OFFICE
PEARL HARBOR, T. H.
1952

HAWAII NATURE NOTES

THE PUBLICATION OF THE
NATURALIST DIVISION, HAWAII NATIONAL PARK
AND THE HAWAII NATURAL HISTORY ASSOCIATION

VOL. V

JUNE 1952

NO. 1

Ferns of Hawaii National Park

by
DOUGLASS H. HUBBARD
Park Naturalist ^[1]



SPECIAL ISSUE

•

PRICE 50 CENTS

[1]

Mr. Hubbard is now Associate Park Naturalist at Yosemite National Park

2



Stately tree ferns border a road in Hawaii National Park

INTRODUCTION

Ferns are among the most interesting plants in the plant world. Varieties in Hawaii range from dainty filmy ferns less than one inch in length to stately tree ferns over 40 feet high, and inhabit areas varying from dry, barren lava flows to impenetrable rain jungles.

More primitive than flowering plants, ferns reproduce by means of small, dust-like bodies called spores. These spores are often carried in the air, and are so lightweight that they can be blown for thousands of miles by winds. The ferns of Hawaii, or their ancestors, apparently reached these islands with the help of winds, except for those introduced recently by man.

HAWAII NATIONAL PARK

Hawaii National Park belongs to the American people. When it was set aside by Congress in 1916 the three volcanoes, Haleakala, Mauna Loa, and Kilauea were of principal interest. It has become increasingly apparent that the beautiful forests of native trees, the birds which depend upon Hawaiian plants for survival, and the magnificent fern jungles are of equal importance and more in need of protection. Outside of the park, trees such as the *koa* (Hawaiian mahogany) and the *ohia* are being cut for lumber, and tree ferns, upon whose trunks orchids are grown, are rapidly being taken for this expanding industry. The National Park Service has been given the responsibility of keeping its areas in as nearly an original condition as possible—a little bit of America, unspoiled. Hawaii National Park—your park—is among the most outstanding of these. Please help us keep it this way.

THIS BOOKLET

This booklet is not intended to be a comprehensive treatment of all the ferns—some 70 species—which have been found to date in Hawaii National Park. Rather it is an illustrated guide to the more abundant species which may be seen by the visitor without wandering far from the beaten path. Abundance has been used as the basis for arrangement, instead of scientific classification, and illustrations take the place of technical descriptions. The more serious observers are referred to the selected bibliography near the end of this publication.

Some disagreement exists today among botanists as to the proper scientific names for certain Hawaiian plant species, and many ferns have been undergoing reclassification. For this reason all of the scientific names that could be found for the various ferns in this booklet have been listed on pages [37](#), [38](#) and [39](#).

A brief code has been included to indicate the type of habitat in which the ferns normally may be found: D=dry, M=moist, D&M=dry and moist. Numbers in bold-face type are used for that particular fern throughout the publication. The page numbers refer to the location of the text and the scientific name for that species.

One inch squares are used for background scale in many of the illustrations.

All photographs are by the author unless otherwise credited.

TREE FERNS

The five species of Hawaiian tree ferns recognized by botanists will appear quite similar to most visitors. The [illustrations below](#) show several of the differences. Most common of the tree ferns are the two species called HAPUU (10, 11) which do not have the coarse bristles of the HAPU III (9) which sometimes grow to a height of 35 feet, including the fronds. The fourth species, the MEU (12), is not common, and is found only on the island of Hawaii. A fifth species is frequent in the highlands of Kauai.

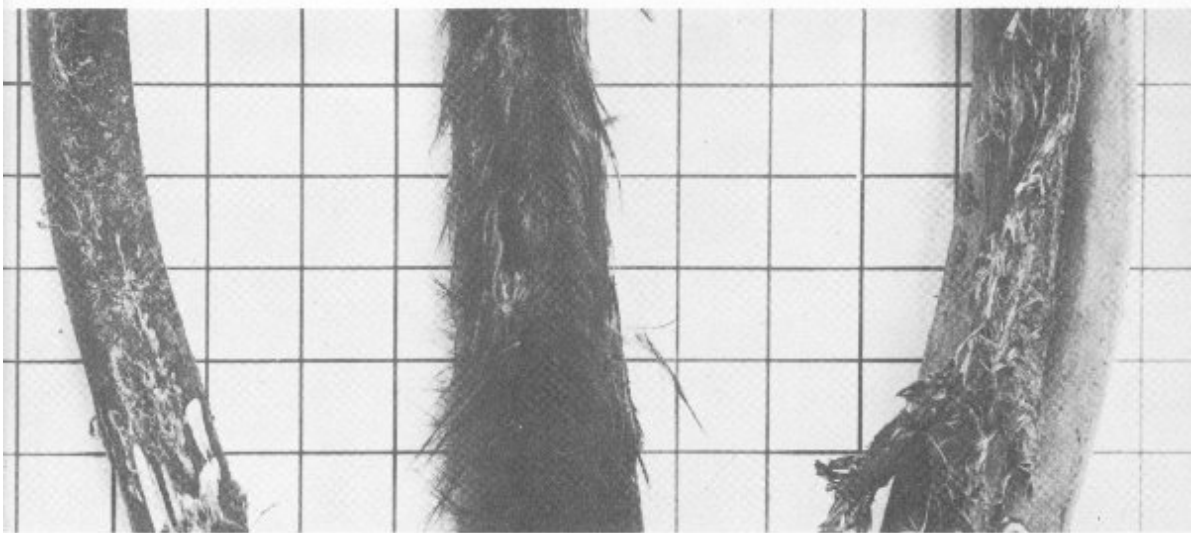
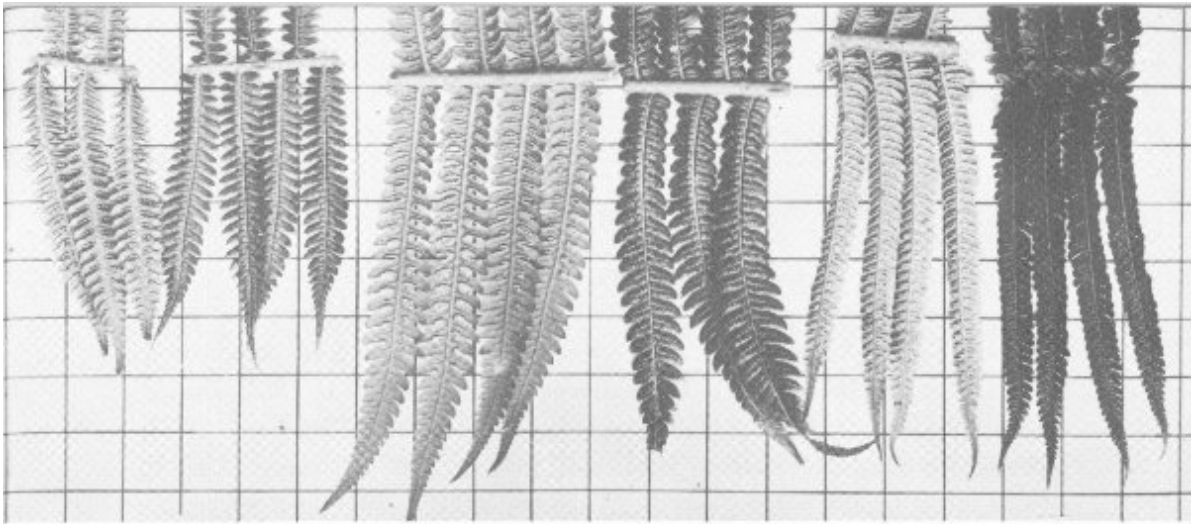
The reproductive bodies of the tree ferns are borne in small, bead-like capsules on the margins of the leaf segments (see [illustration below](#)). These capsules, which open by means of small lids, may be responsible for the generic name of the ferns, *Cibotium*, from a Greek word meaning a small vessel.

The buds of these ferns are covered with the silky PULU, discussed below. According to Hillebrand (5, p. 546) ferns of the genus *Cibotium* range over Mexico and Central America, the Philippine Islands, Sumatra, South China and India. It is an interesting speculation as to which of these places provided the spores which gave the Hawaiian tree ferns their beginning.

Much of what appears to be the trunk on these and many other ferns is in reality a useful mass of aerial roots which absorb moisture and protect the tree trunk from damage. These roots also offer a fertile place for seeds of many other plants to take root and grow. So common is it to find ohia trees growing on tree fern trunks that the ferns are often called *mother of ohia*. Unappreciative of the support given them in their early growth the trees often send roots to earth which become so large that the fern finally is killed. Much starch is present in the trunks of tree ferns. This was resorted to by the Hawaiians for food in time of famine. Young fronds of the tree ferns are used in preparing several exotic dishes today, and many Park ferns have suffered severely by having the new growth removed for this purpose.



Multi-branched fronds characterize tree-ferns



FronD segments, segment closeups, and frond stalk sections of MEU (12) (left), HAPU III (9) (center), and HAPUU (11) tree ferns.

PULU

The young fronds of certain ferns, especially tree ferns, are covered with a bronze-colored silky floss called “pulu.” Used by the ancient Hawaiians as an absorbent, for surgical dressings, and even in embalming their dead, pulu became an item of export importance to the mainland from the 1860’s to 1884. During this period some 4,271,119 pounds were shipped, according to Degener (2). It was used primarily for stuffing mattresses, pillows, and upholstery. Carl P. Russell in his *One-Hundred Years in Yosemite* (1947, p. 102) quotes Caroline M. Churchill, a visitor to Leidig’s, a Yosemite hostelry established in 1869: “At this place the beds are cleanly and wholesome, although consisting of pulu mattresses placed upon slat bedsteads.”

The remains of the old pulu factory may be seen today in Kilauea section about 1.8 miles below Makaopuhi Crater on the trail to Napau Crater. Here the pulu was dried and baled before being taken seaward over the pali to Keauhou Landing for loading onto mainland-bound vessels.

Although a high percentage of pulu was gathered from Kilauea ferns, the industry fortunately terminated about 1885, and no more ferns were cut for the small amount of pulu obtained from each. The hike to the pulu factory is not strenuous, and is one of the most interesting in Hawaii National Park.



{Pulu.}

AMAUMAU AND AMAU

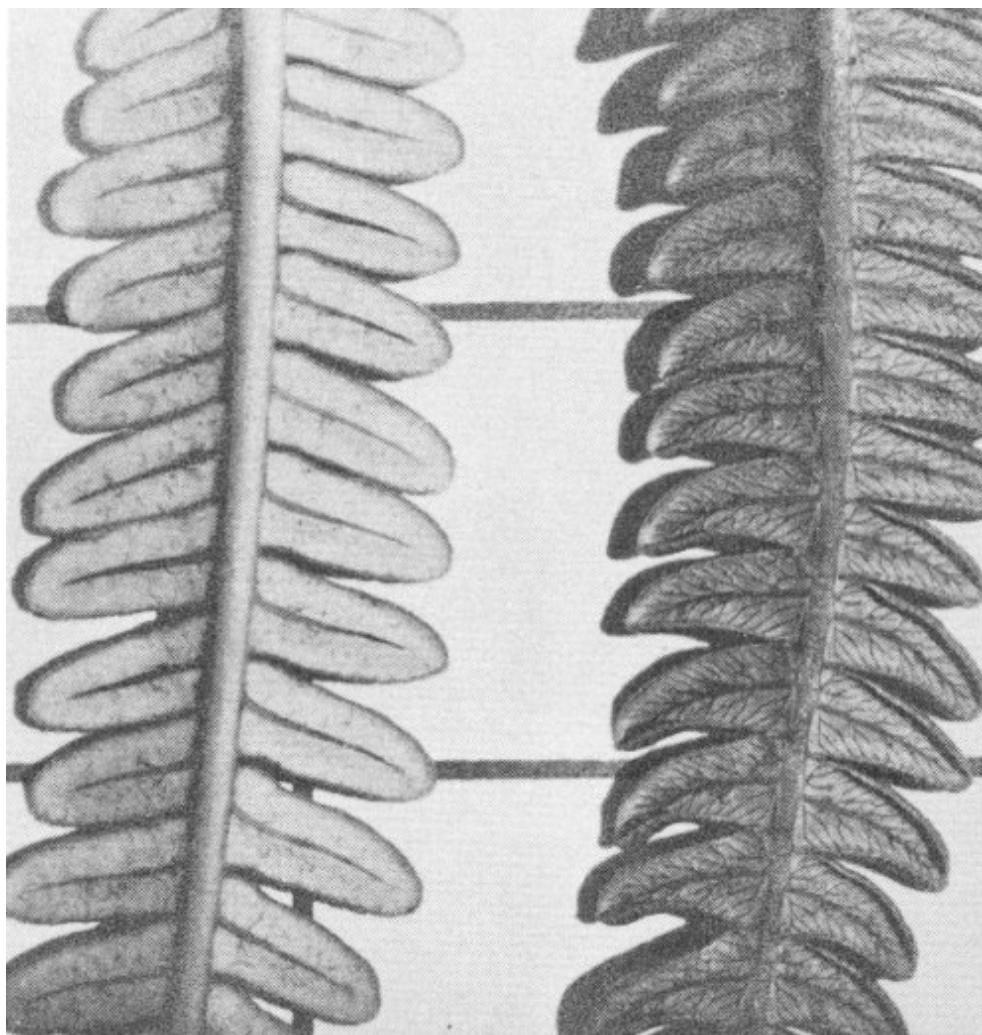
The most abundant ferns growing beneath the larger tree ferns are AMAUMAU (40) and the AMAU (41). Found frequently in both the Kilauea and Haleakala sections of the Park, the amaumau is usually larger than the less common amau and the fronds are not as hairy. The easiest method of identification, however, is by comparing the undersides of the frond segments (see [illustration](#)). The amaumau is smooth while the amau is prominently veined. Usually smaller than the true tree ferns, and known only from the Hawaiian Islands, ferns of this genus may reach heights as great as 12 feet. A comparison of the photographs of these ferns with the tree ferns will show that the entire frond of the AMAU or AMAUMAU resembles a single segment of the many-branched frond of the tree fern.

The AMAUMAU fern has a high resistance to heat and drought, and may often be seen growing on the Kau Desert near Kilauea crater. It is one of the first plants to take root on new lava flows, in company with the ohia tree. The [illustration below](#) shows a specimen growing on the 1877 lava which forms the floor of Keanakokoi crater.

Fowler (4, p. 16) states that the Hawaiians prepared a red dye used on KAPA cloth from the outer part of the trunk of the amaumau. Leaves of the amau provided a lining and thatching for houses, and the young leaves and pith of the stems were occasionally cooked for food.



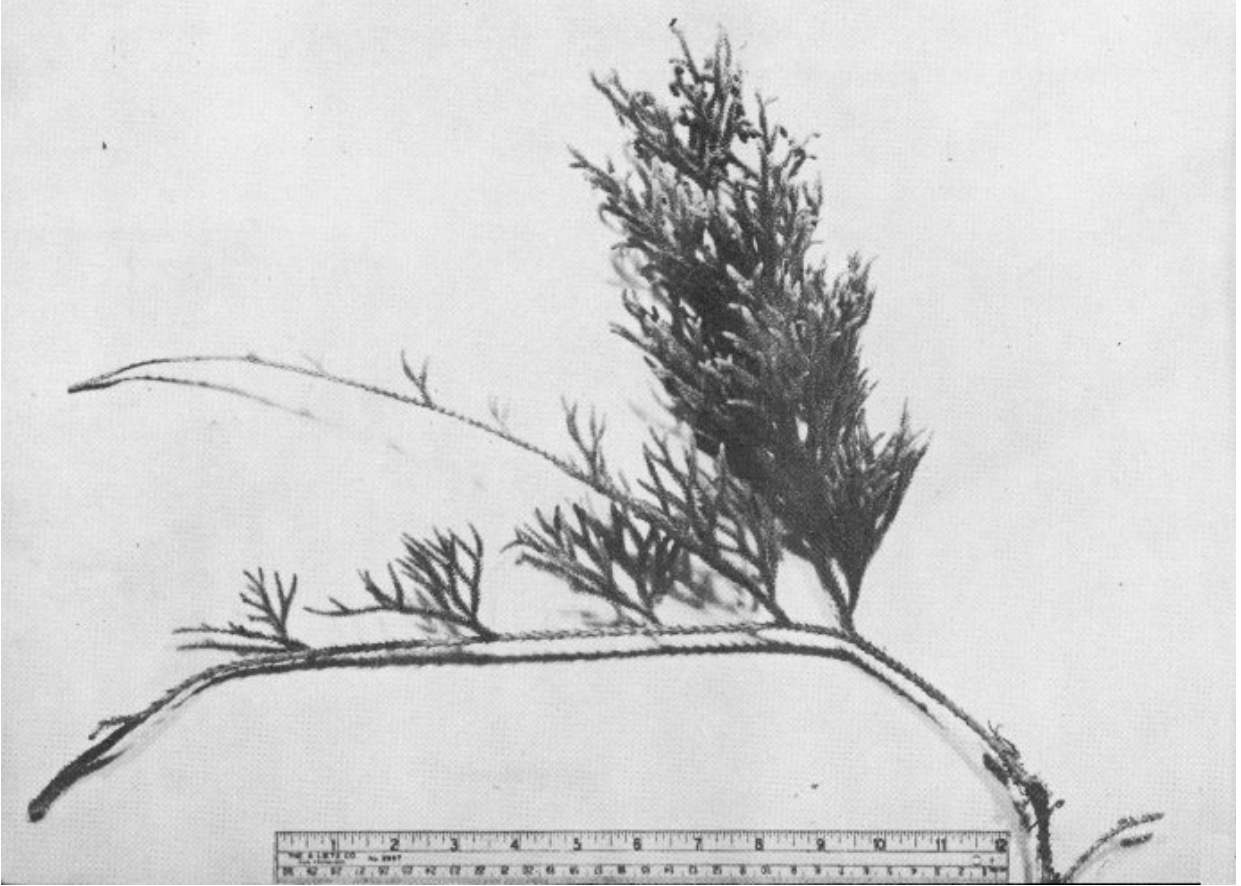
{AMAUMAU *growing on lava.*}



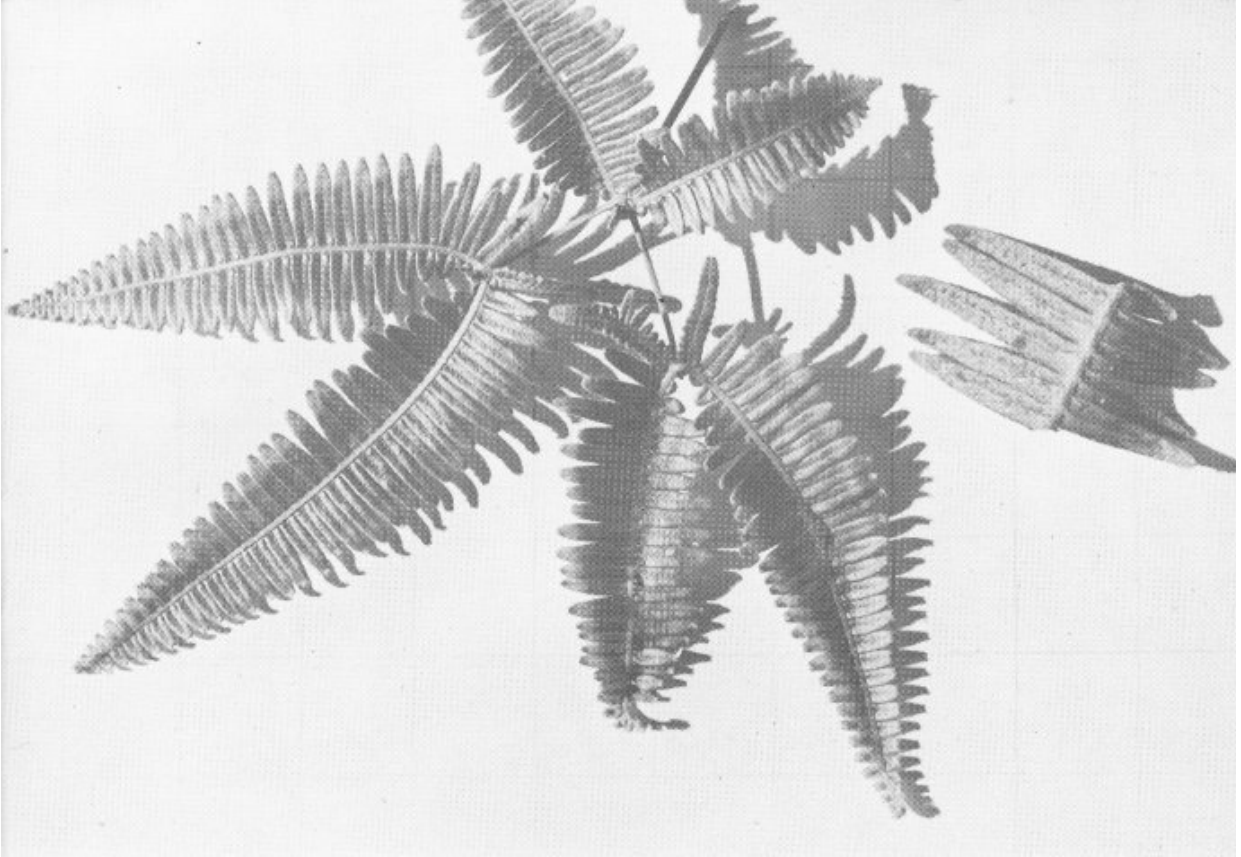
{Underside of frond segments of AMAUMAU and AMAU.}



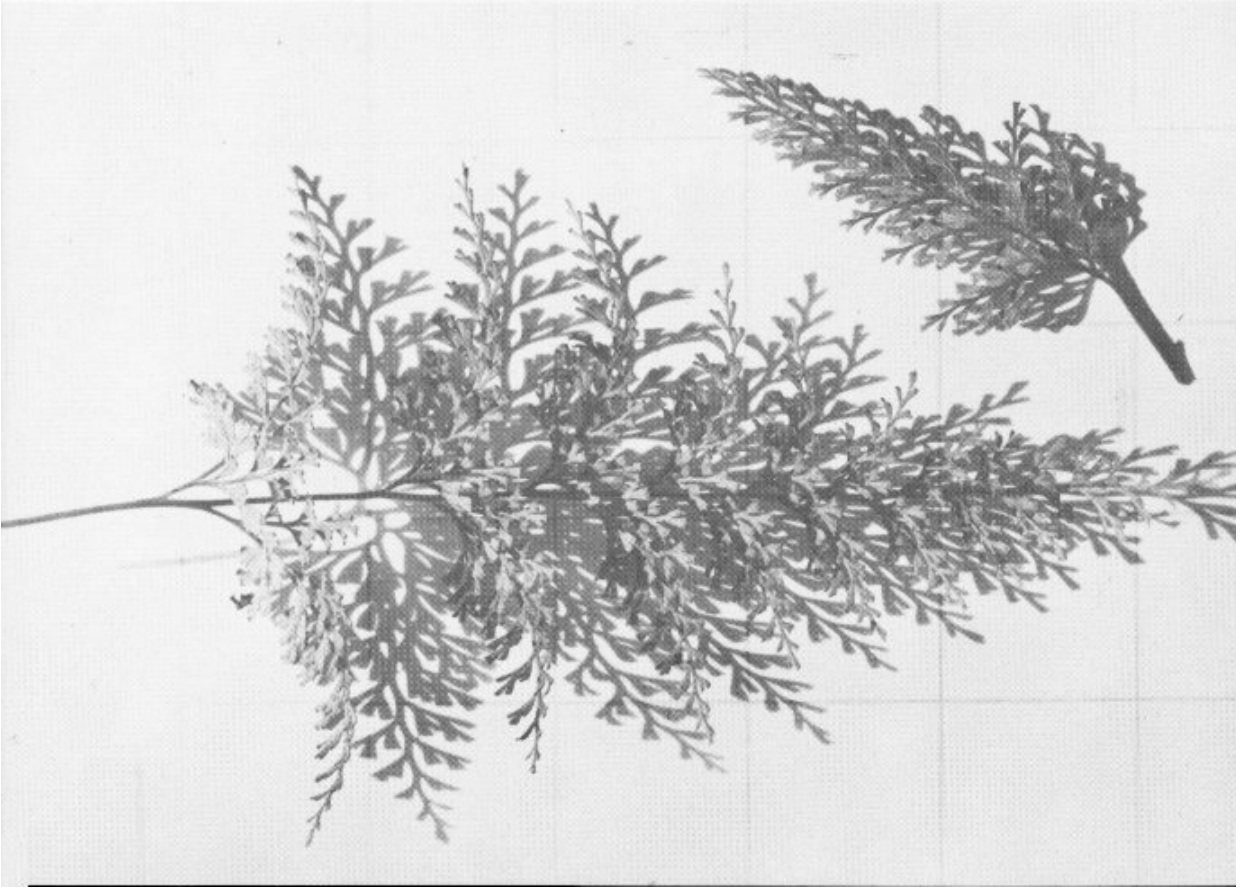
In the rain forest near the old Pulu Factory. At least six different kind of ferns may be seen in this photograph.



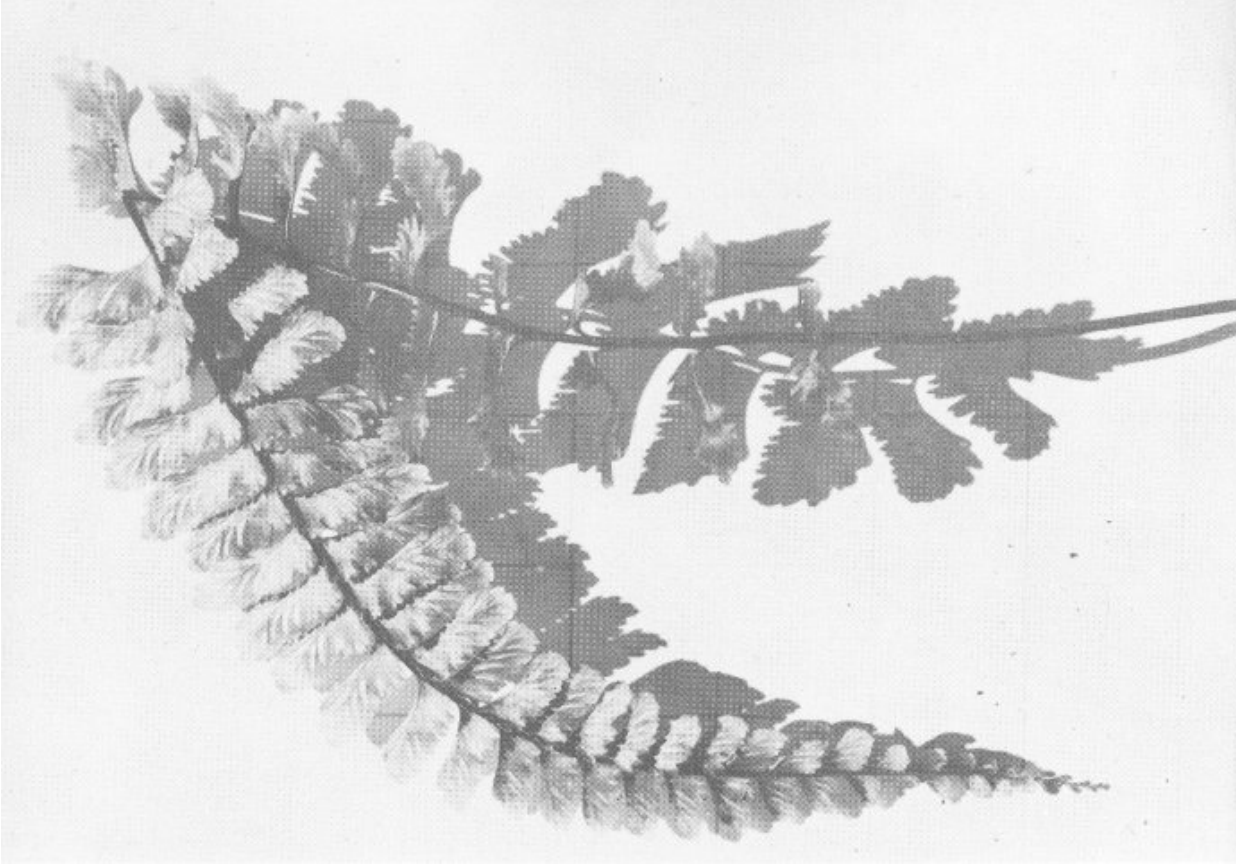
WAWAEIOLE or Clubmoss. D&M, (58-61) (pages [36](#) and [39](#))



ULUHE or False Staghorn. D&M, (4) (pages [32](#) and [37](#))



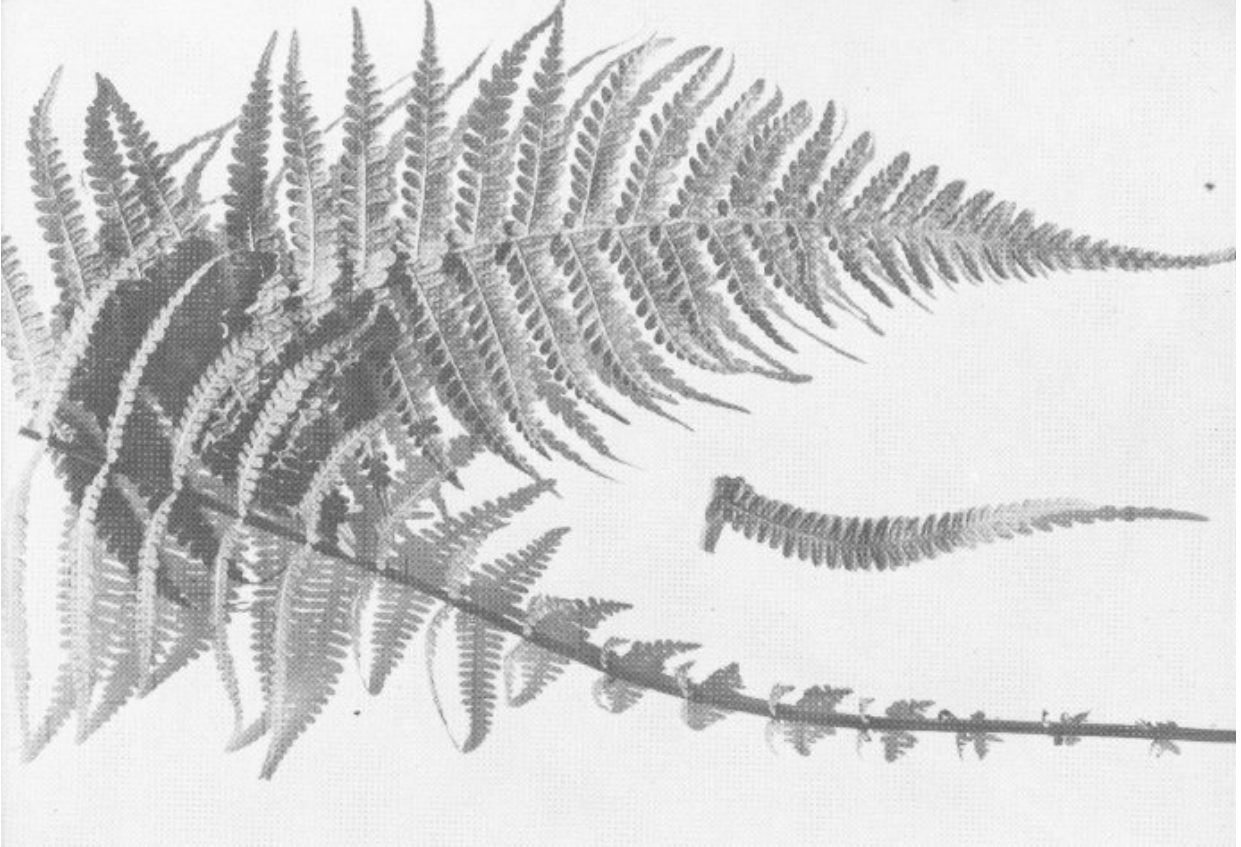
PALAA. D, (14) (pages [32](#) and [37](#))



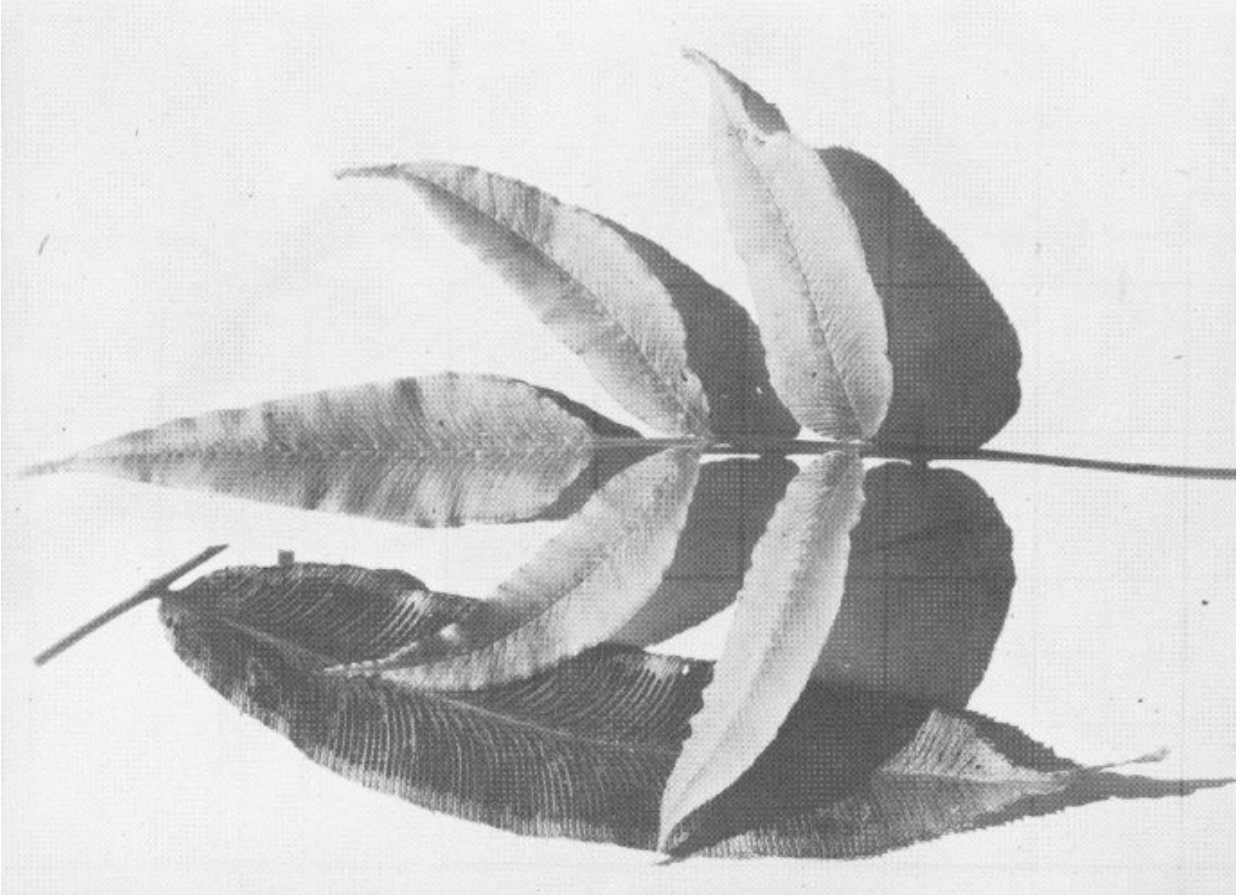
РАМОН. М, (48) (pages [35](#) and [38](#))



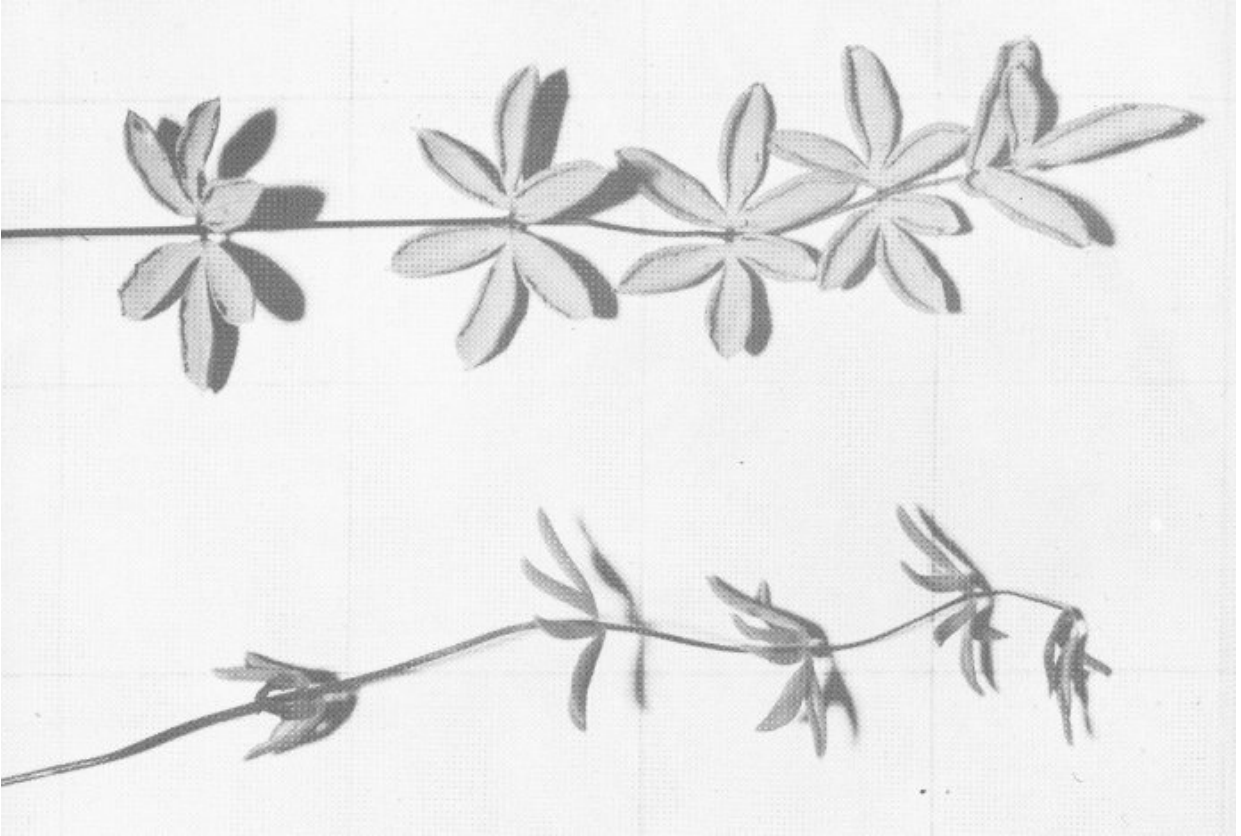
Pamoho at the mouth of the Thurston Lava Tube



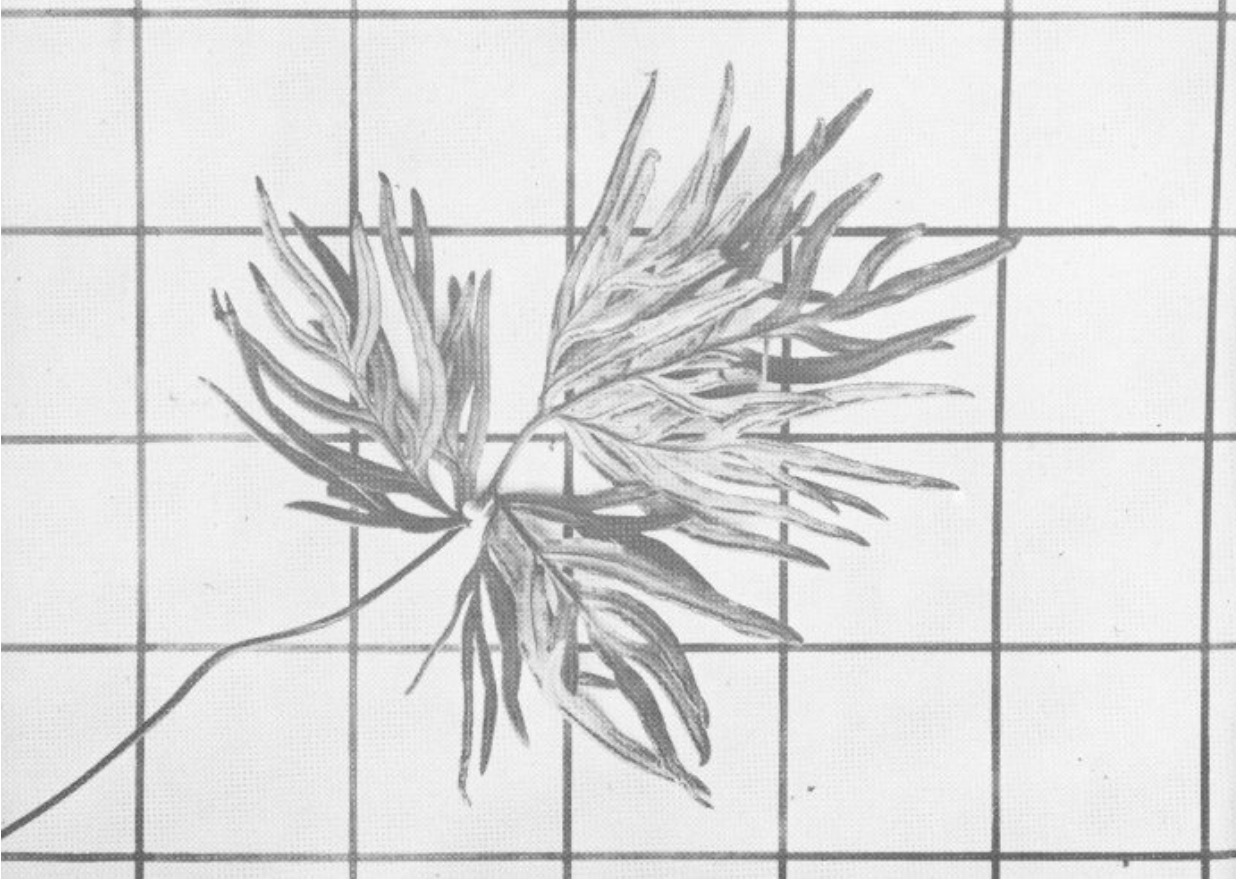
PALAPALAI O KAUMAAPUA. M, (34) (pages [34](#) and [38](#))



LOULU. M, (19) (pages [33](#) and [37](#))



LAUKAHI or Cliffbrake. D, (20) (pages [33](#) and [37](#))



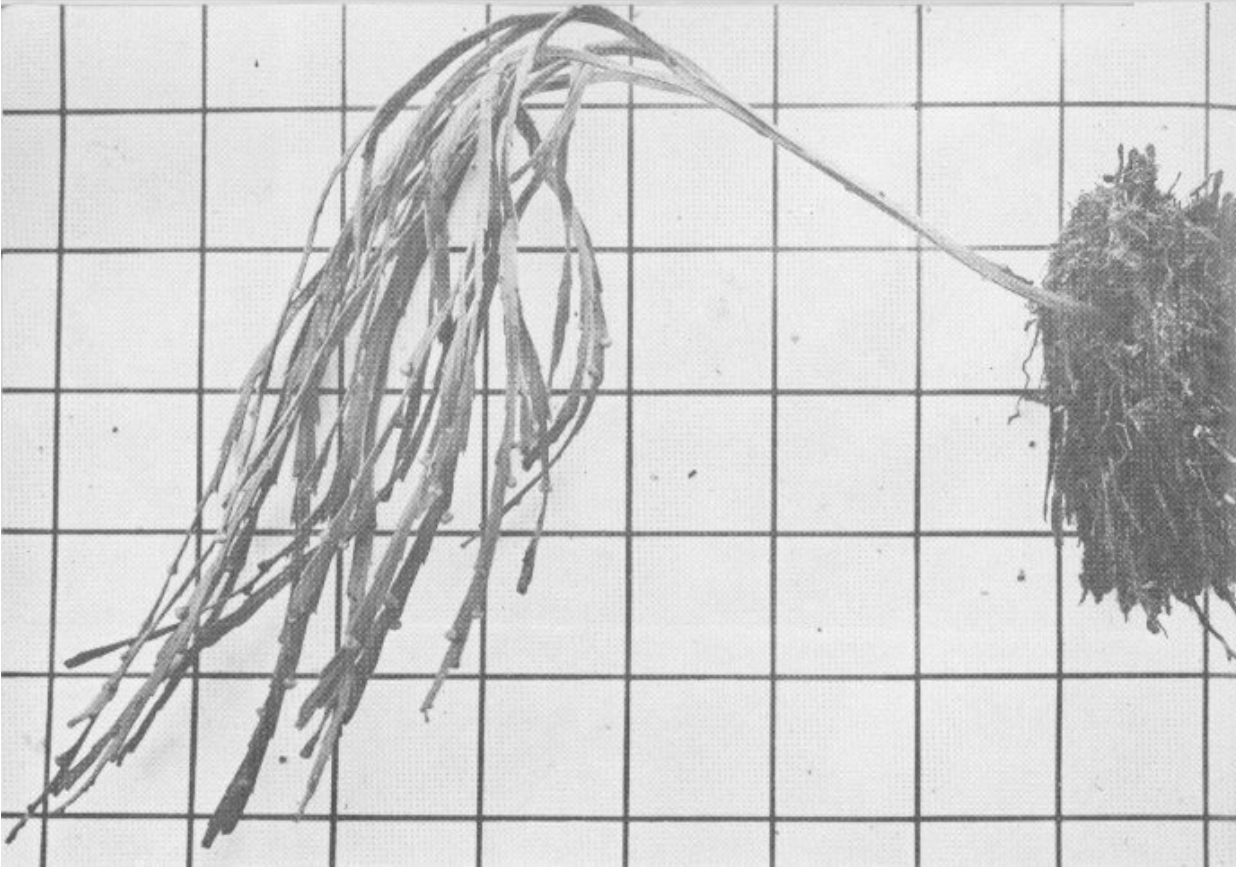
Lance fern or DORYOPTERIS. D, (21) (pages [33](#) and [37](#))



OWALII or Maidenhair Spleenwort. D, (47) (pages [35](#) and [38](#))



KILAU or Bracken. D, (15) (pages [33](#) and [37](#))



PIPL. D, (63) (pages [36](#) and [39](#))

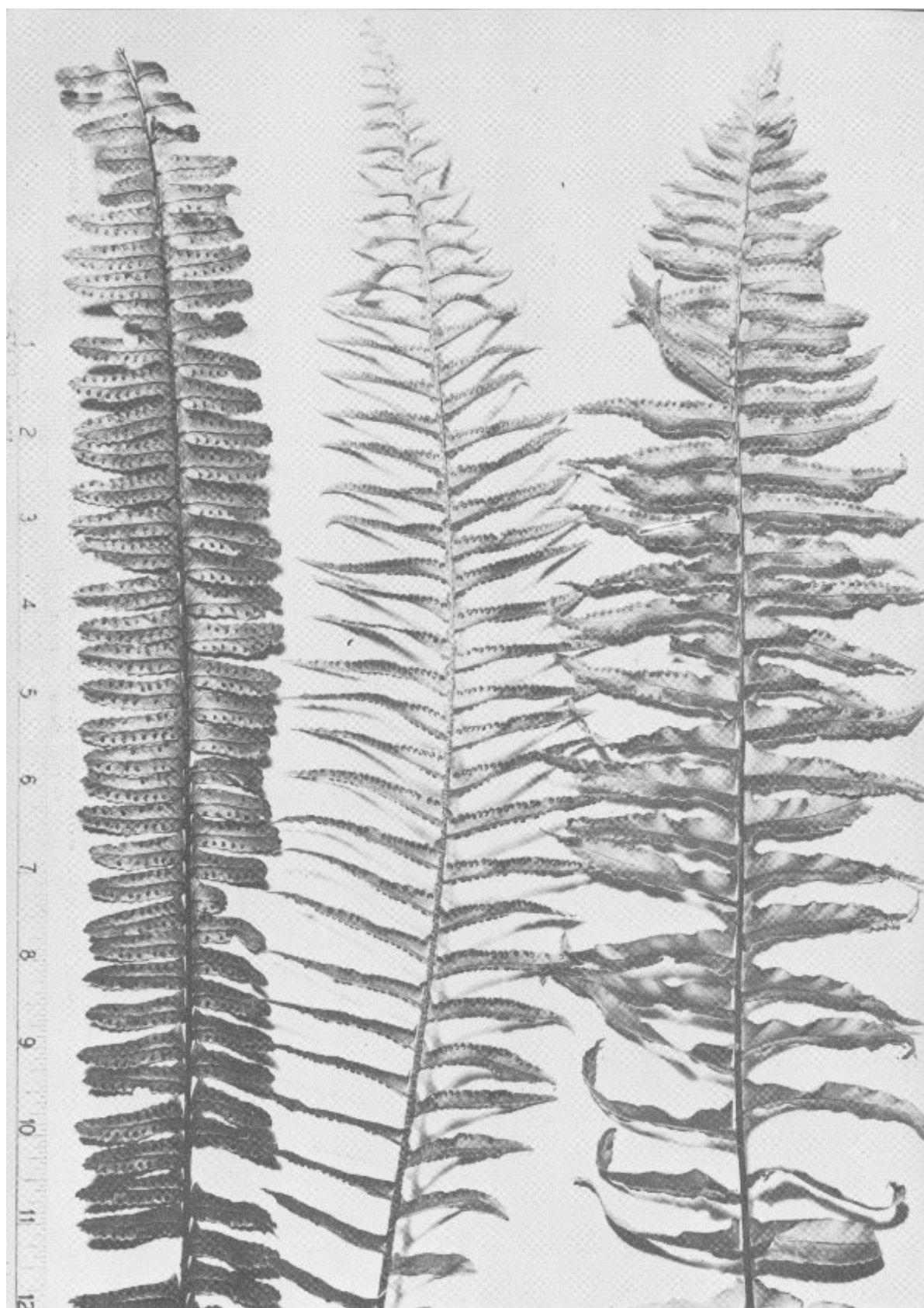


MOA. M, (62) (pages [36](#) and [39](#))



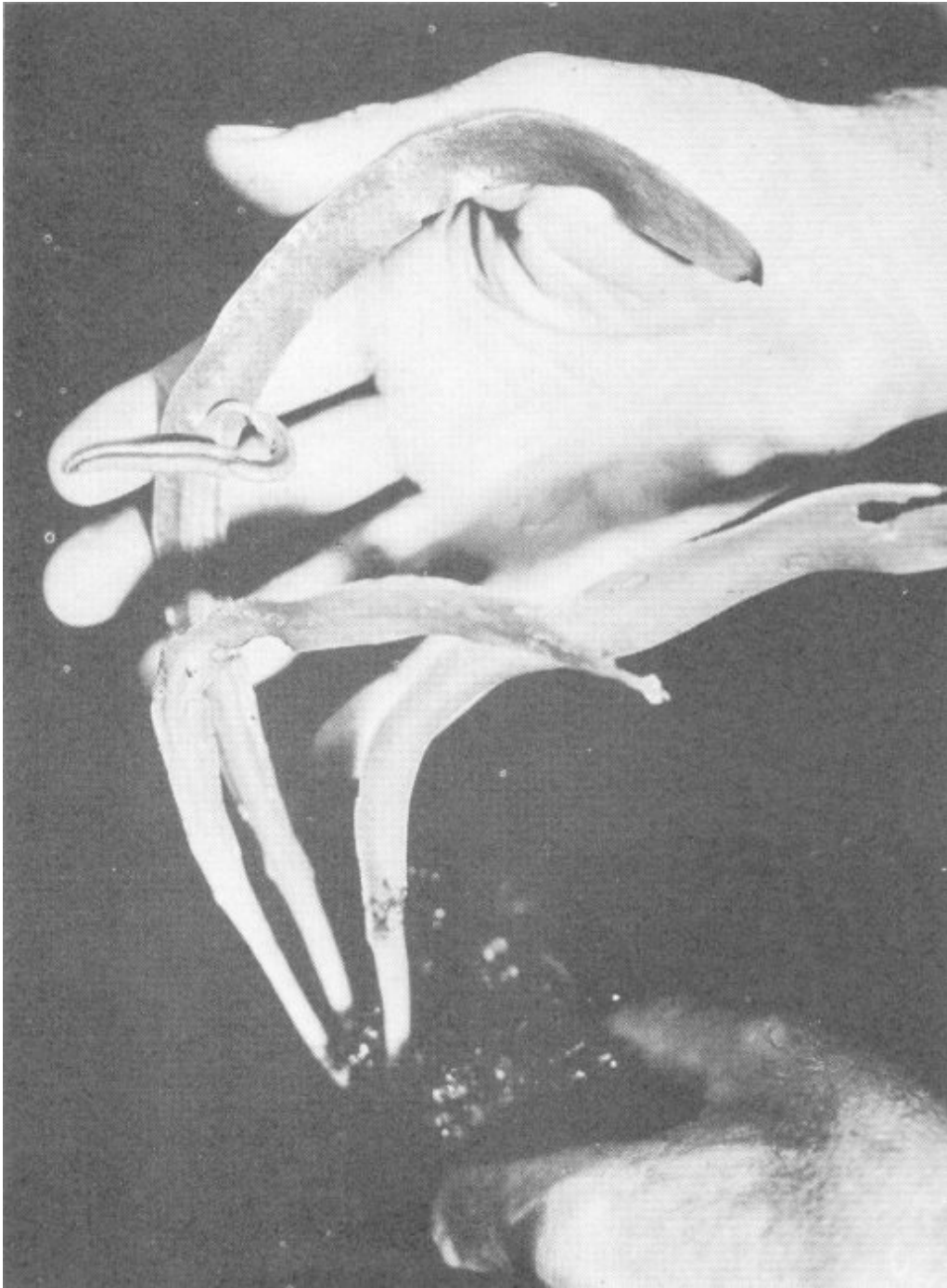
Werner Stoy

Dry-land ferns and grasses form the ground cover at this overlook above the Koolau Gap, with Hanakauhi, “Maker of the Mist”, in the distance. Haleakala Section, Hawaii National Park.

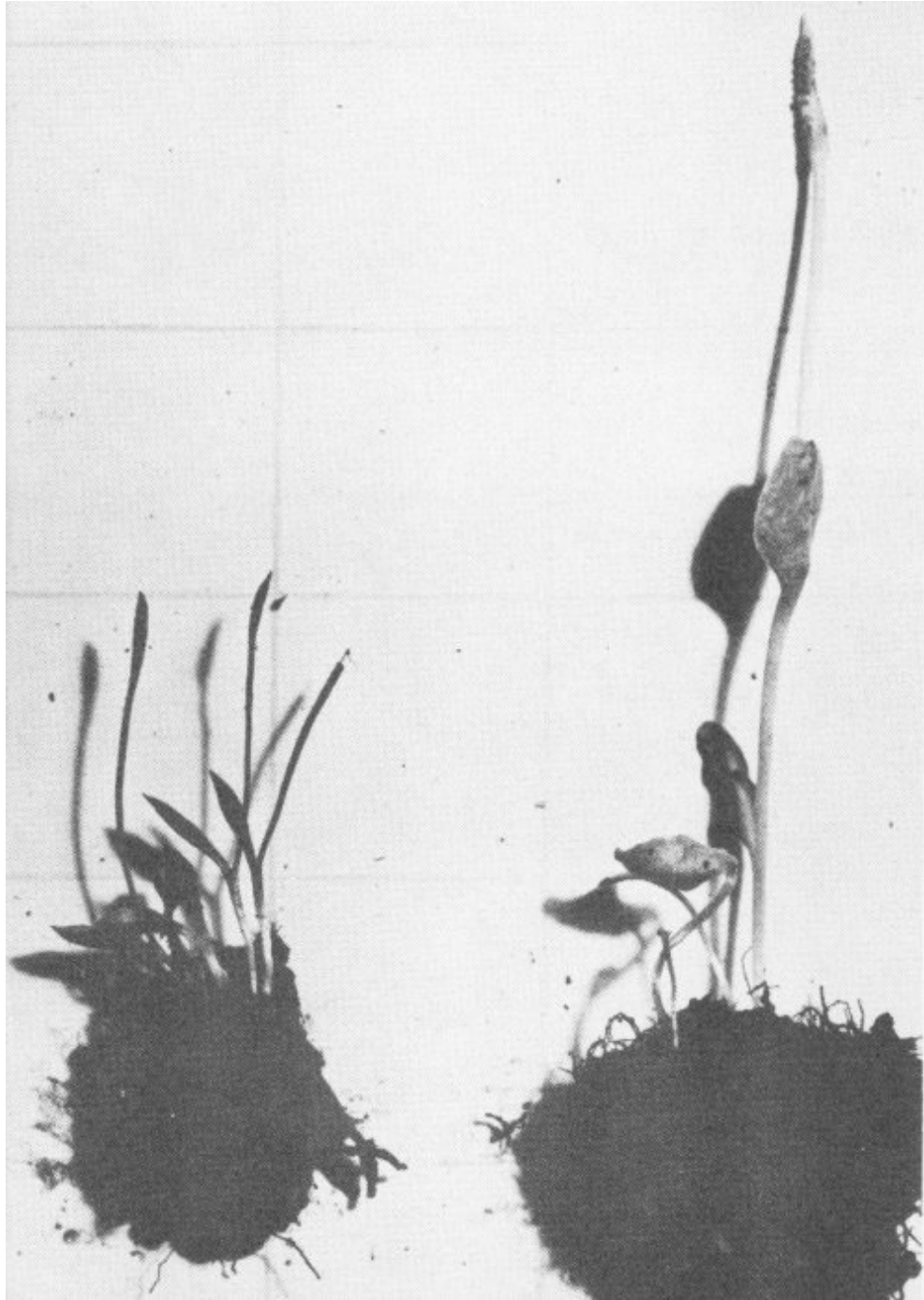


NIANIAU or Sword Ferns. From the left, Narrow Swordfern (25), Scaly Swordfern (26), and Common Swordfern (24).

17



Adder's Tongue. (Pages [32](#) and [37](#))
***Ophioglossum pendulum* (3)**

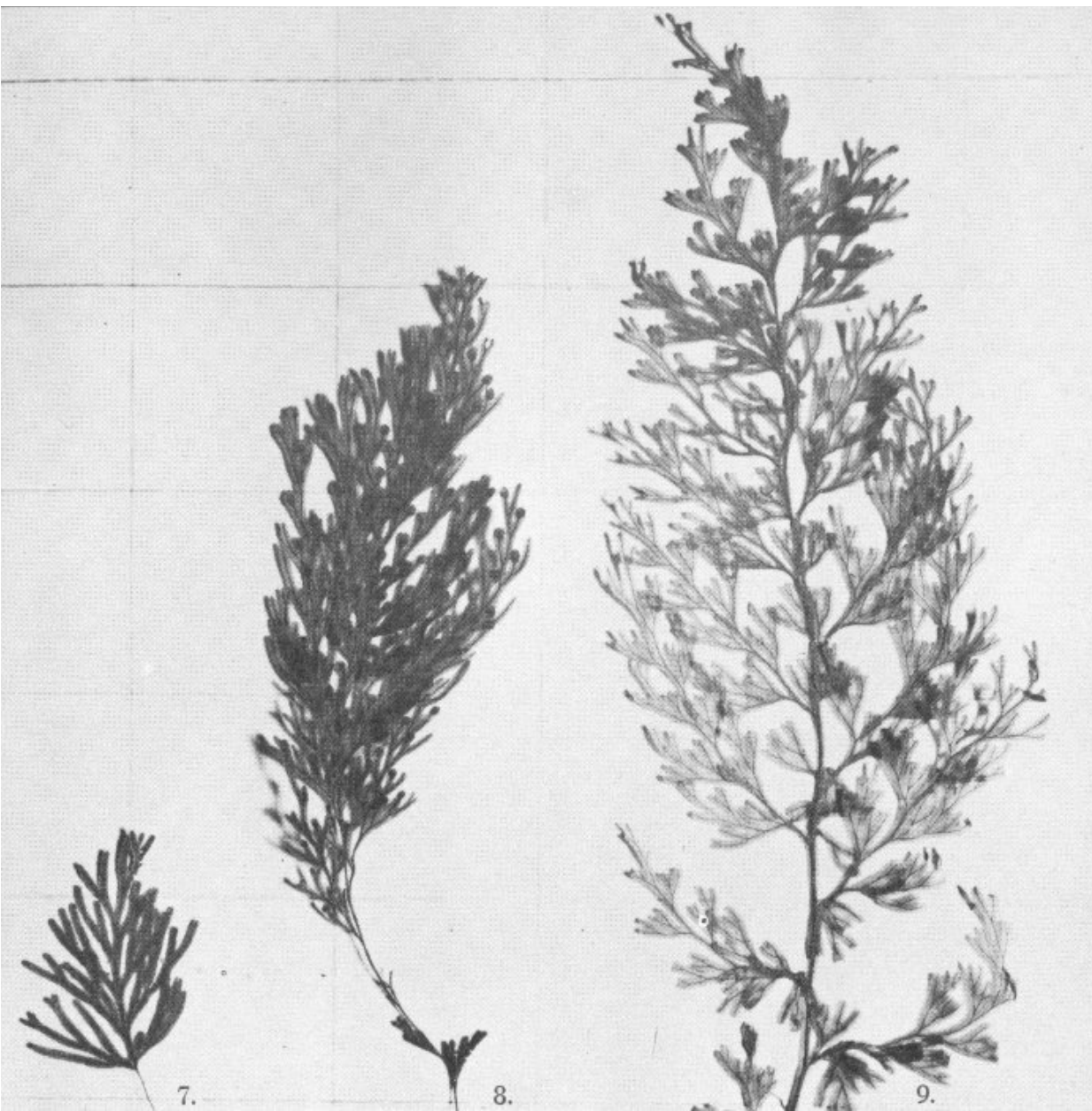


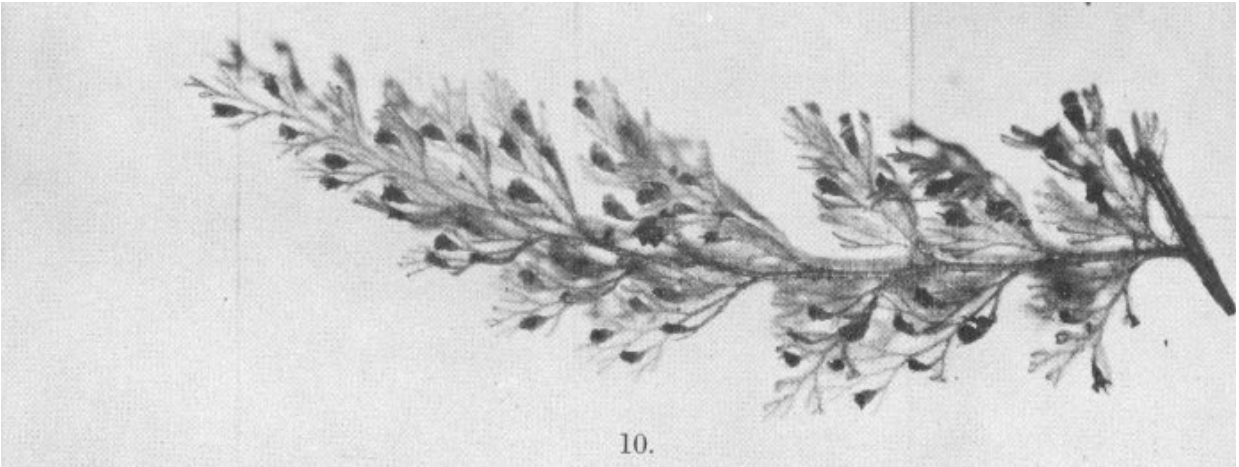
Ophioglossum petiolatum (1) left
Ophioglossum nudicaule (2)



Common epiphytes, or air plants

1. KIHĪ. M, (54) (pages [36](#) and [38](#))
2. PAI. M, (55) (pages [36](#) and [38](#))
3. EKAHA AKOLEA. M, (51) (pages [36](#) and [38](#))
4. GRAMMITIS HOOKERI. M, (52) (pages [36](#) and [38](#))
5. AMPHORADENIUM SARMENTOSUM. M, (56) (pages [36](#) and [38](#))
6. WAHINI NOHO MAUNA. M, (57) (pages [36](#) and [38](#))
7. PALAILAULI. M, (5) (pages [32](#) and [37](#))
8. PALAIHINAHINA. M, (6) (pages [32](#) and [37](#))
9. OHIAKU. M, (7) (pages [32](#) and [37](#))
10. KĪLAU. M, (8) (pages [32](#) and [37](#))

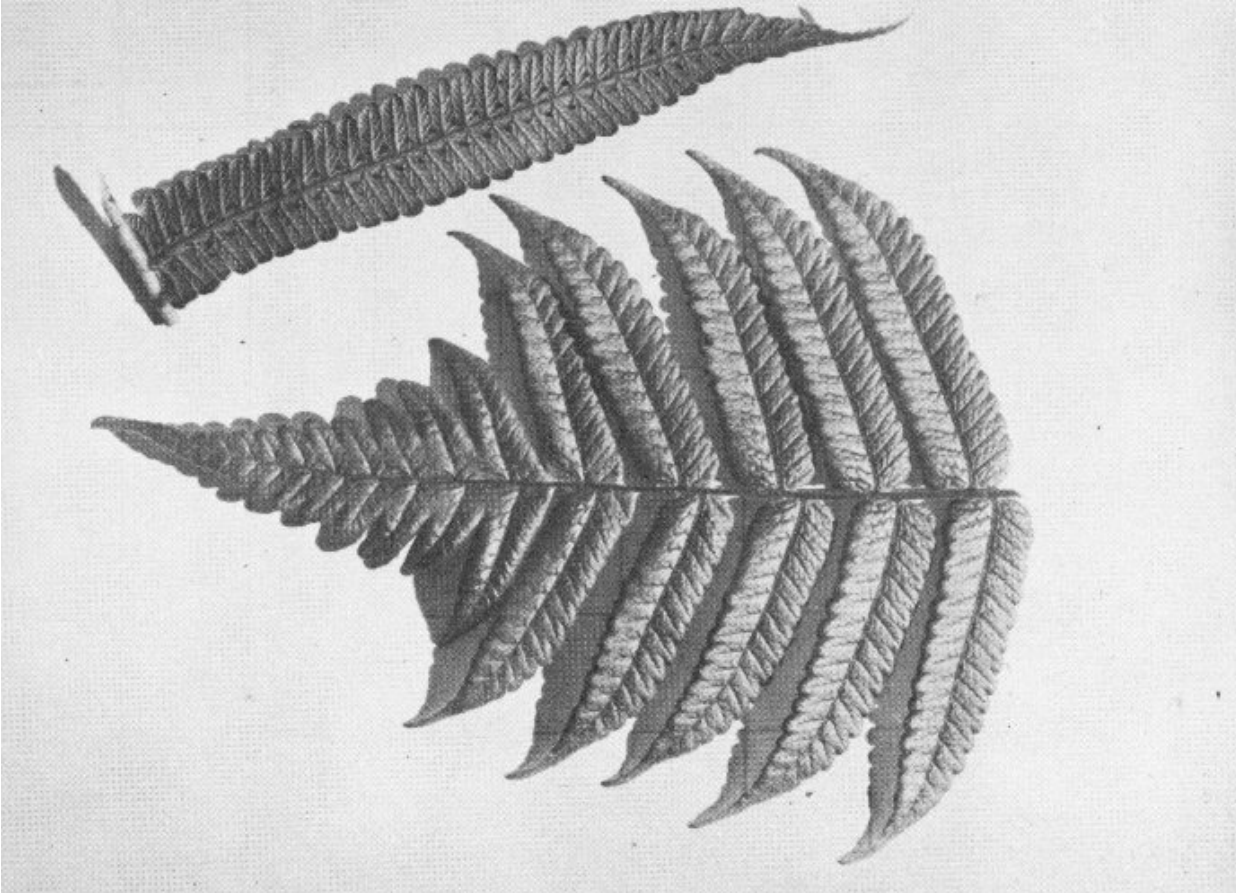




Amaumau ferns form beautiful patterns as they send out new fronds. Kilauea Section, Hawaii National Park.



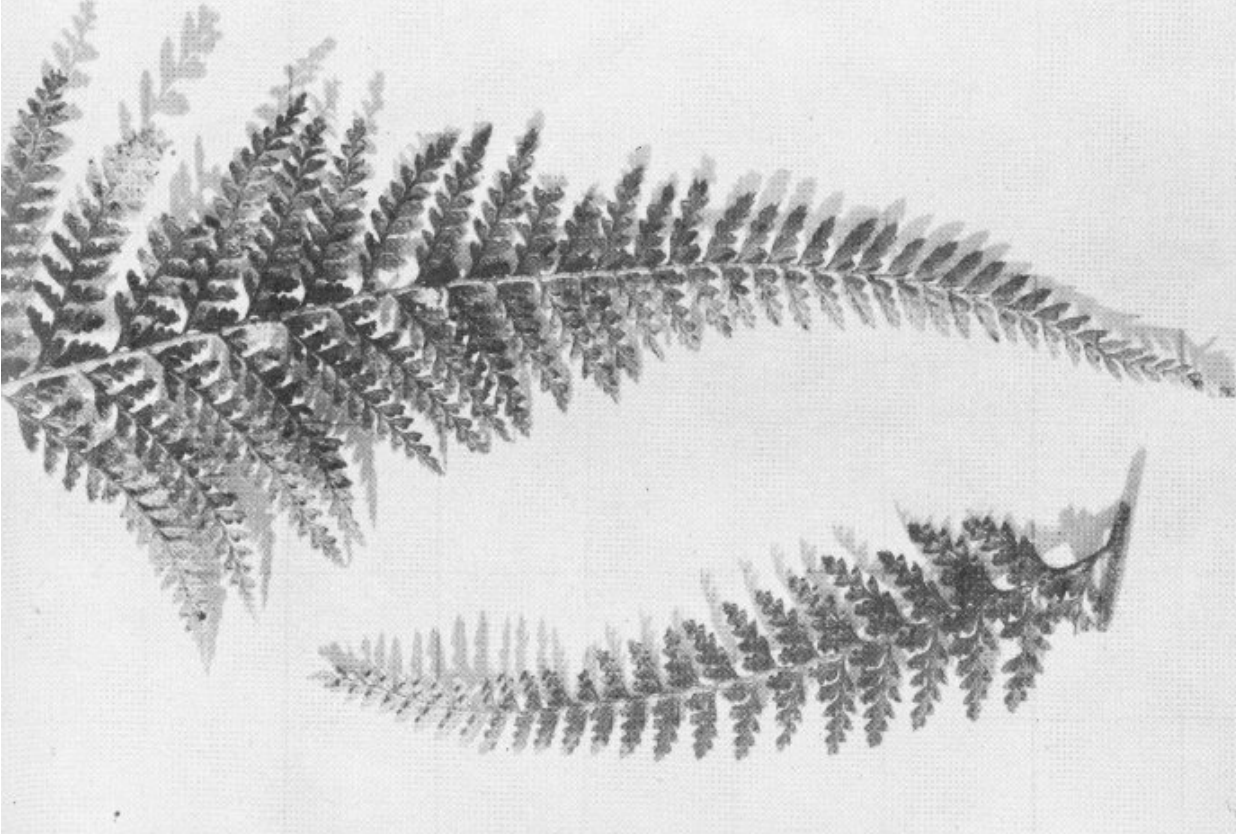
EKAHA or Birdnest Fern. M, (42) (pages [35](#) and [38](#)). The dark areas on undersurfaces are spores.



Cyclosorus sandwicensis. M, (36) (pages [35](#) and [38](#))



The Elephant-Tongue ferns: From the left, *Elaphoglossum reticulatum*, small form (31), *E. aemulum* (28), *E. gorgoneum* (29), *E. reticulatum*, large form (28), and *E. hirtum* (30) (pages [34](#) and [38](#)). All are called “EKAHA.” The dark surfaces are spores.



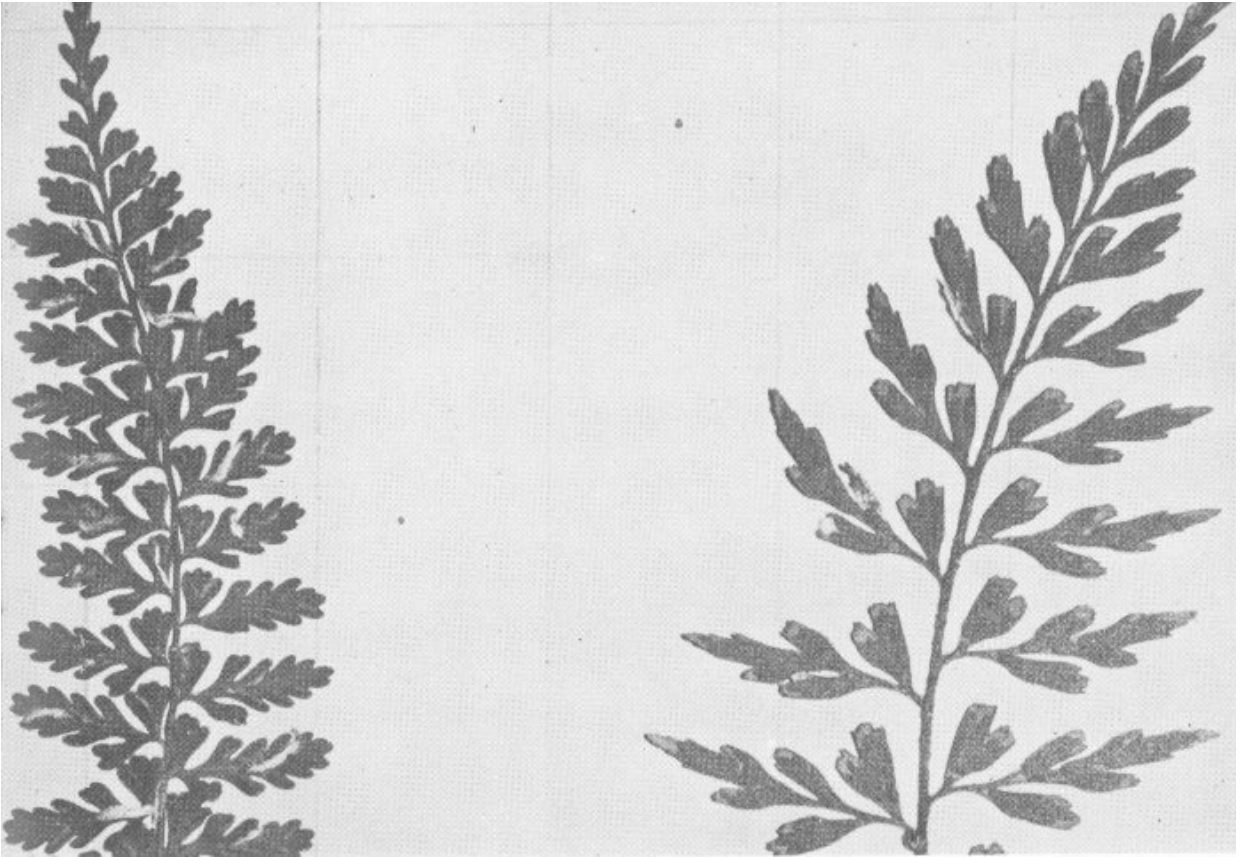
PALAPALAI. D&M, (13) (pages [32](#) and [37](#))



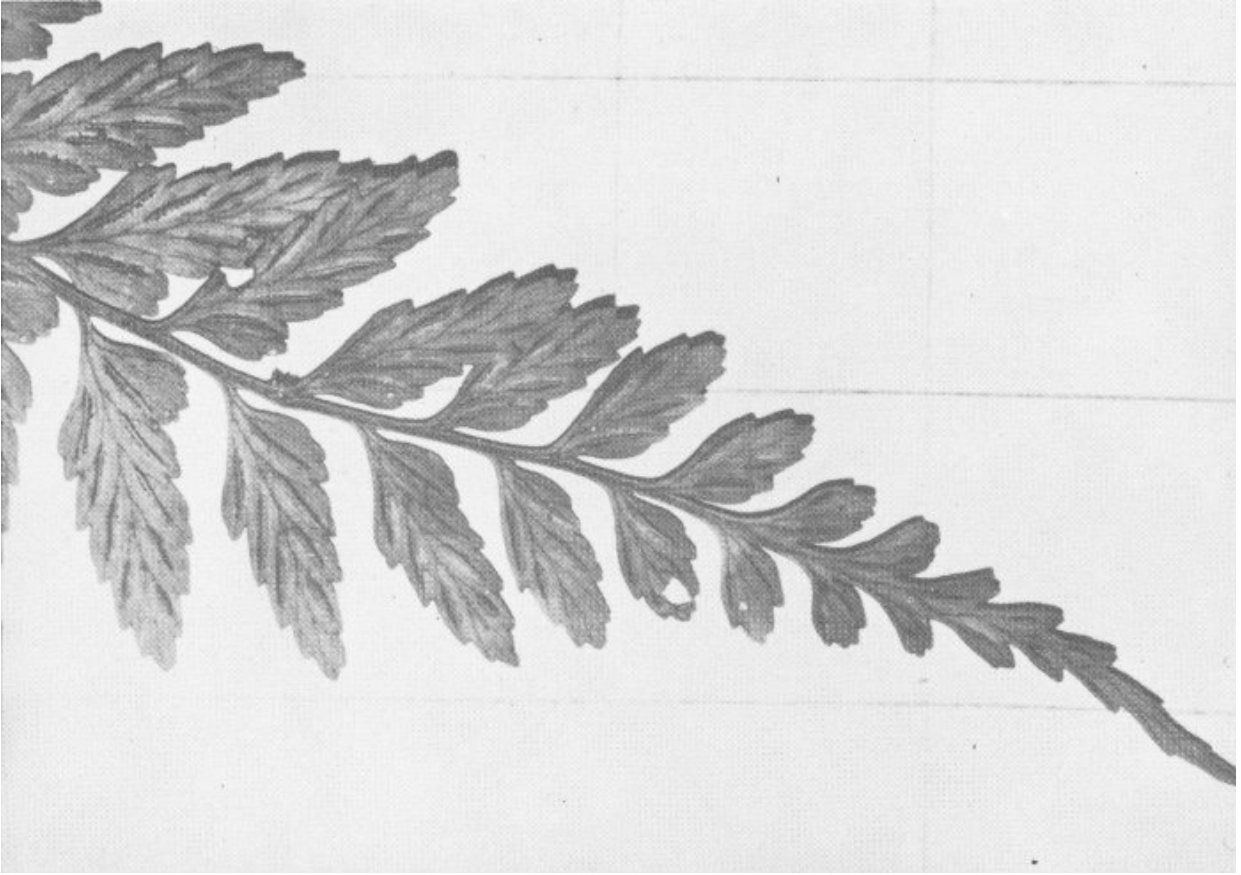
IWAIWA. D, (49) (pages [35](#) and [38](#))



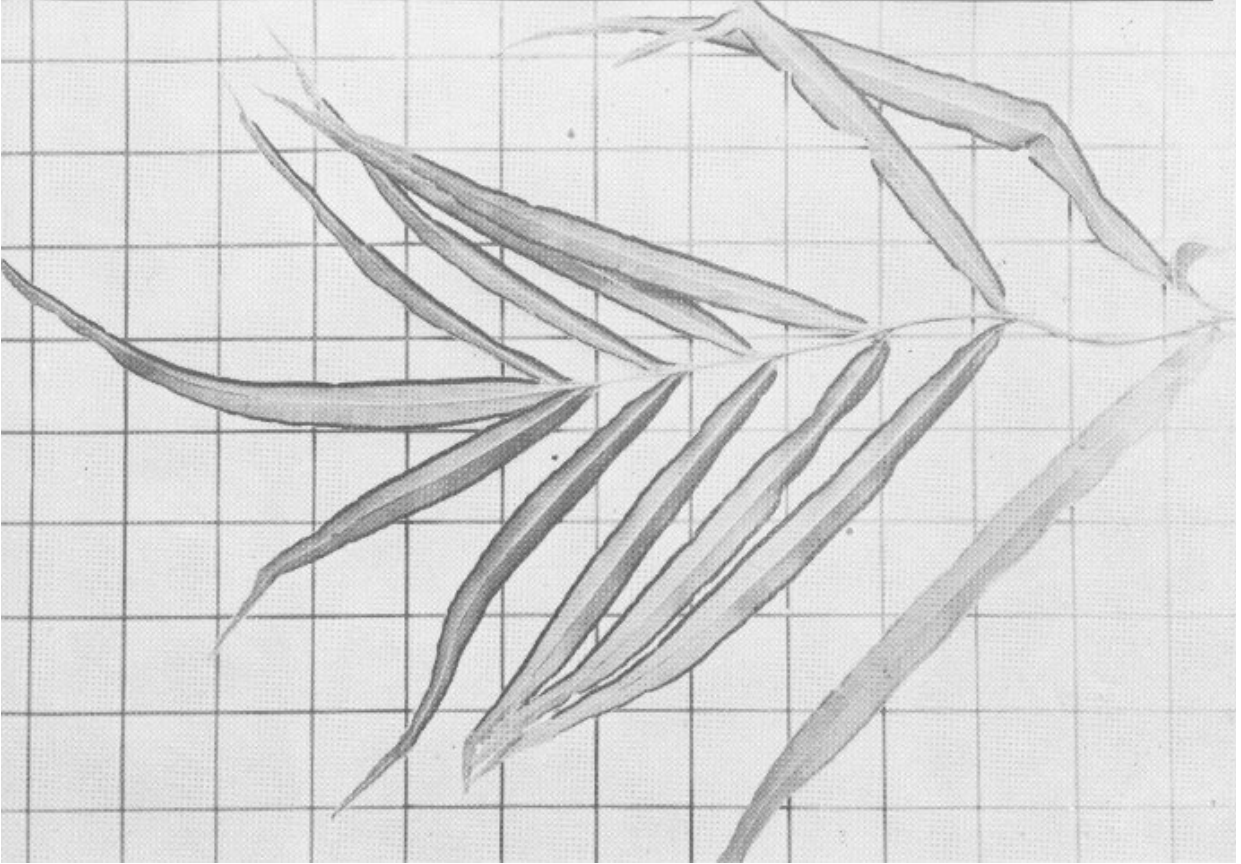
Asplenium contiguum. M, (43) (pages [35](#) and [38](#))



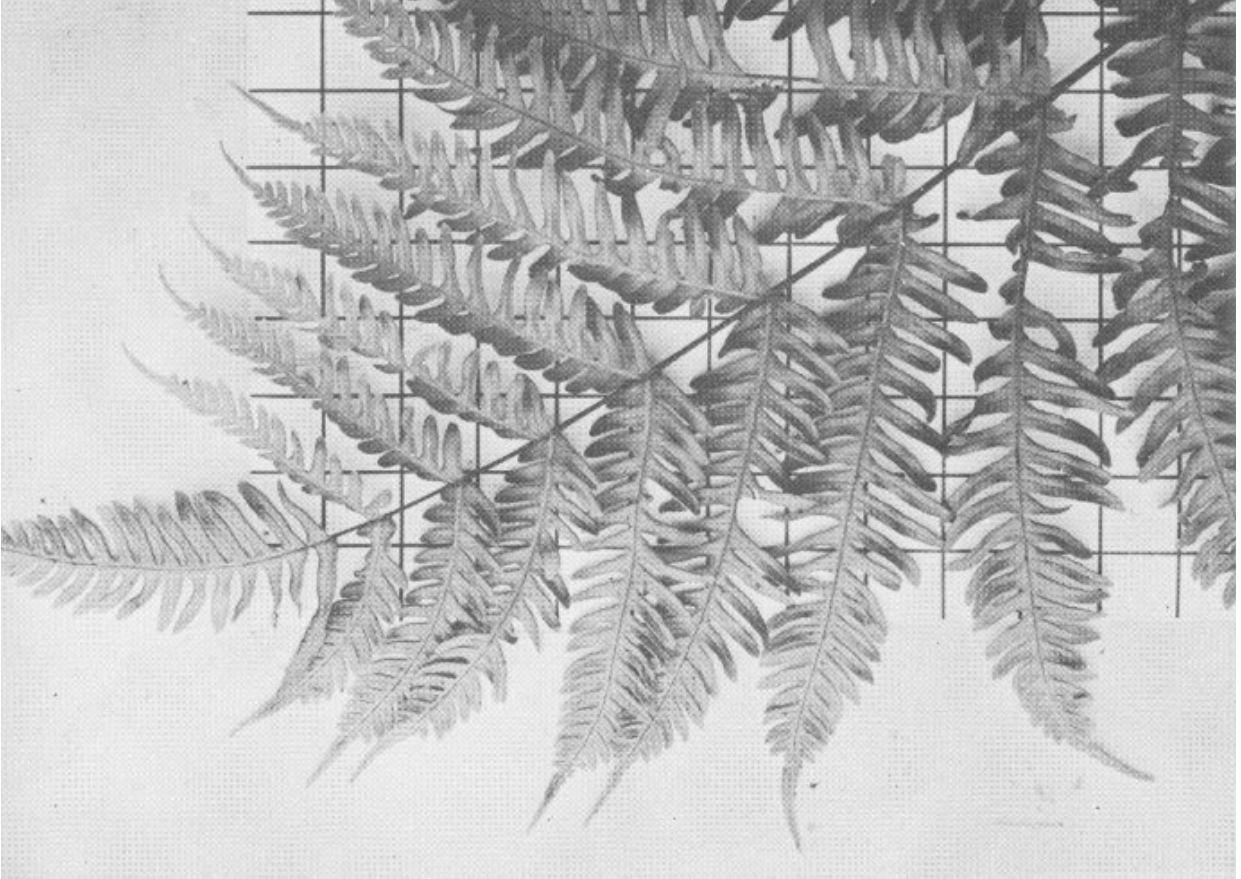
Asplenium Macraei. M, (46) (pages [35](#) and [38](#)). IWAIWA O KANE. D, (44) (pages [35](#) and [38](#))



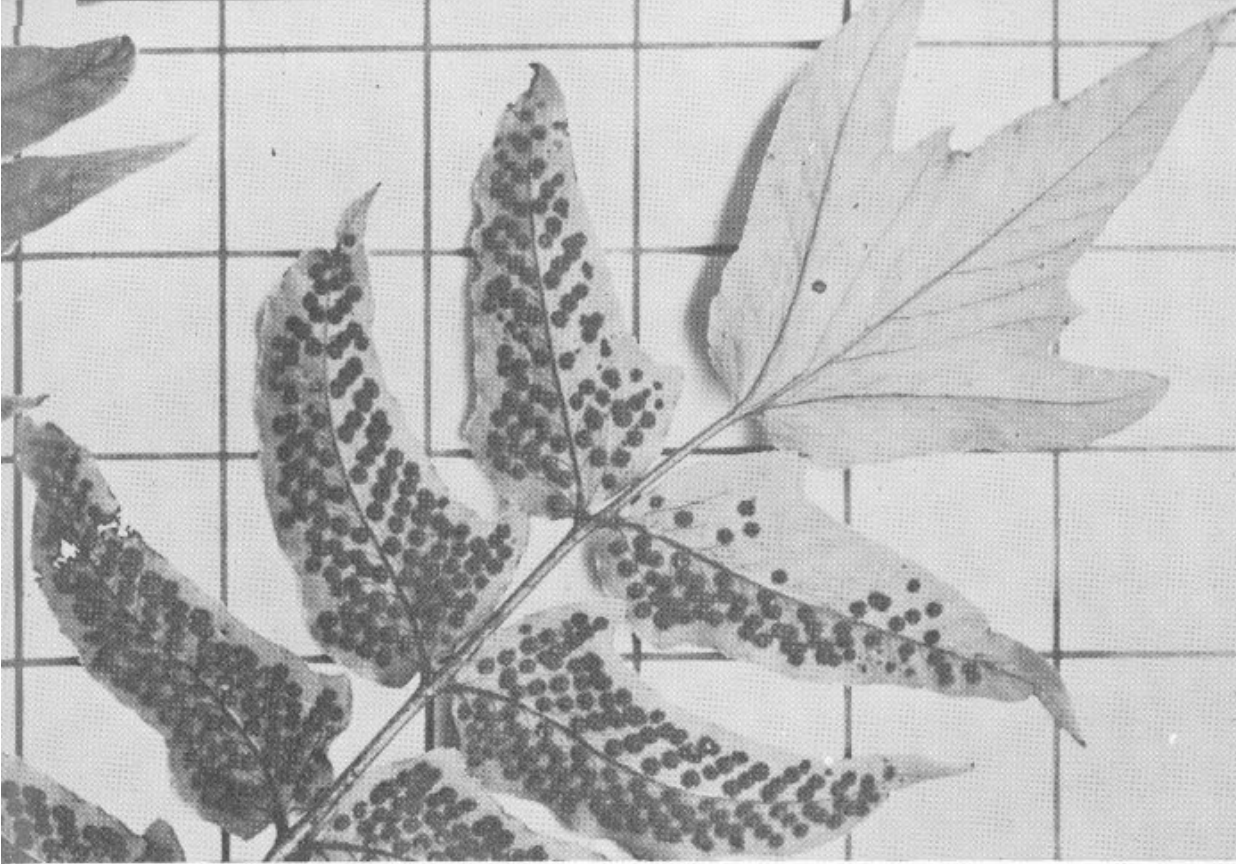
PIIHILOU MANAMANA. M, (45) (pages [35](#) and [38](#))



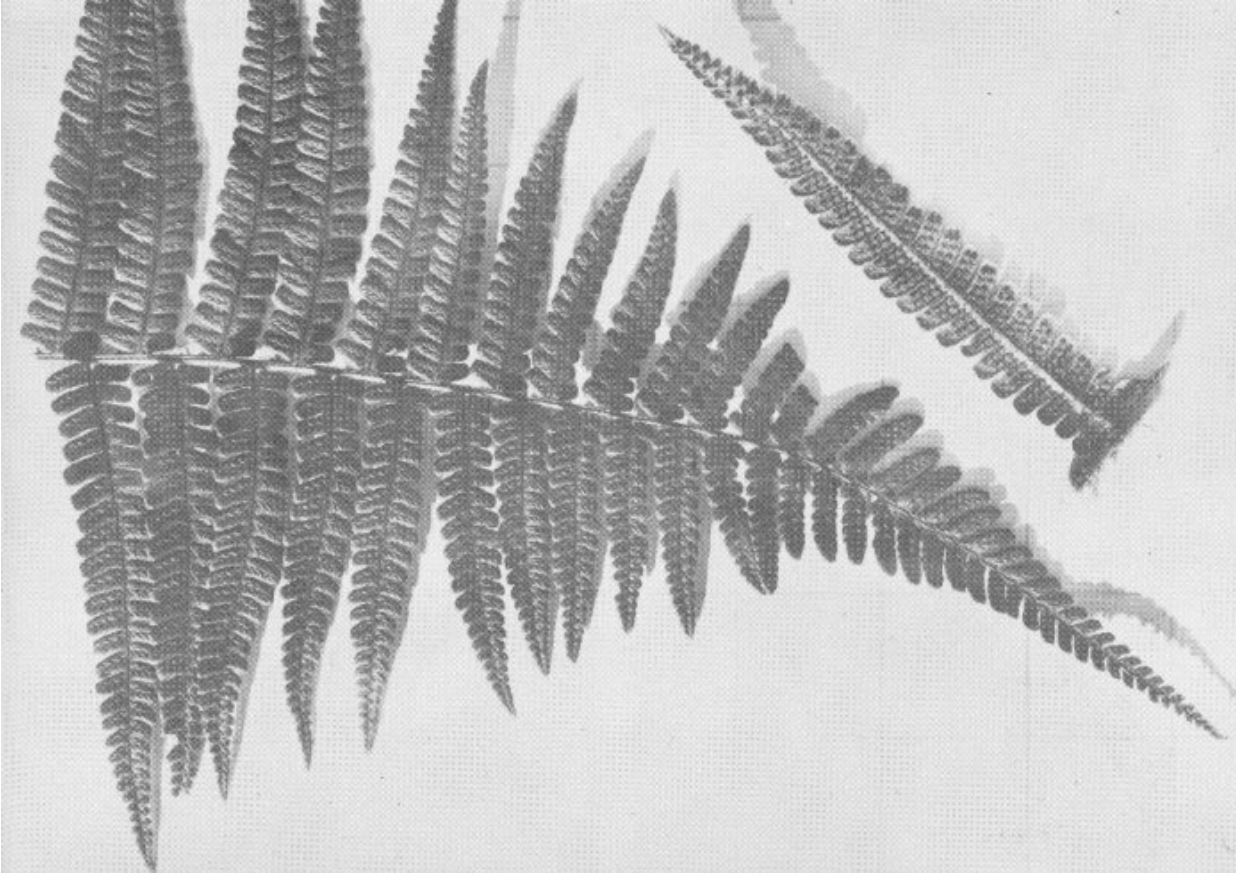
OwALII or Cretan Brake. D&M, (16) (pages [33](#) and [37](#))



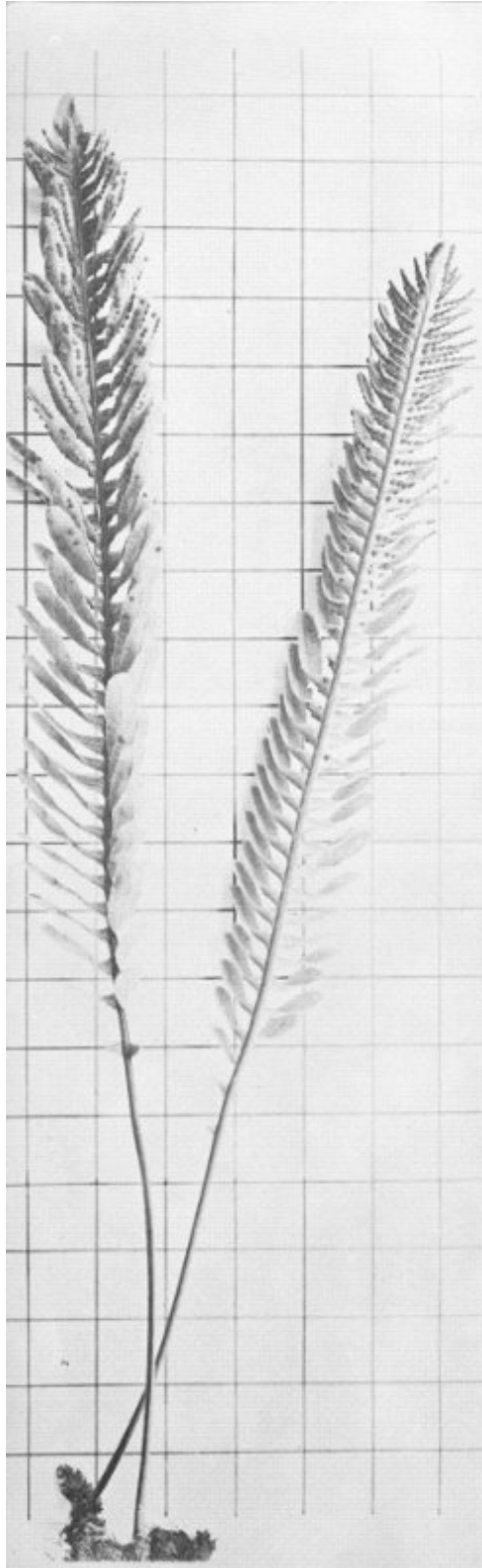
WAIMAKANUI. D, (17) (pages [33](#) and [37](#))



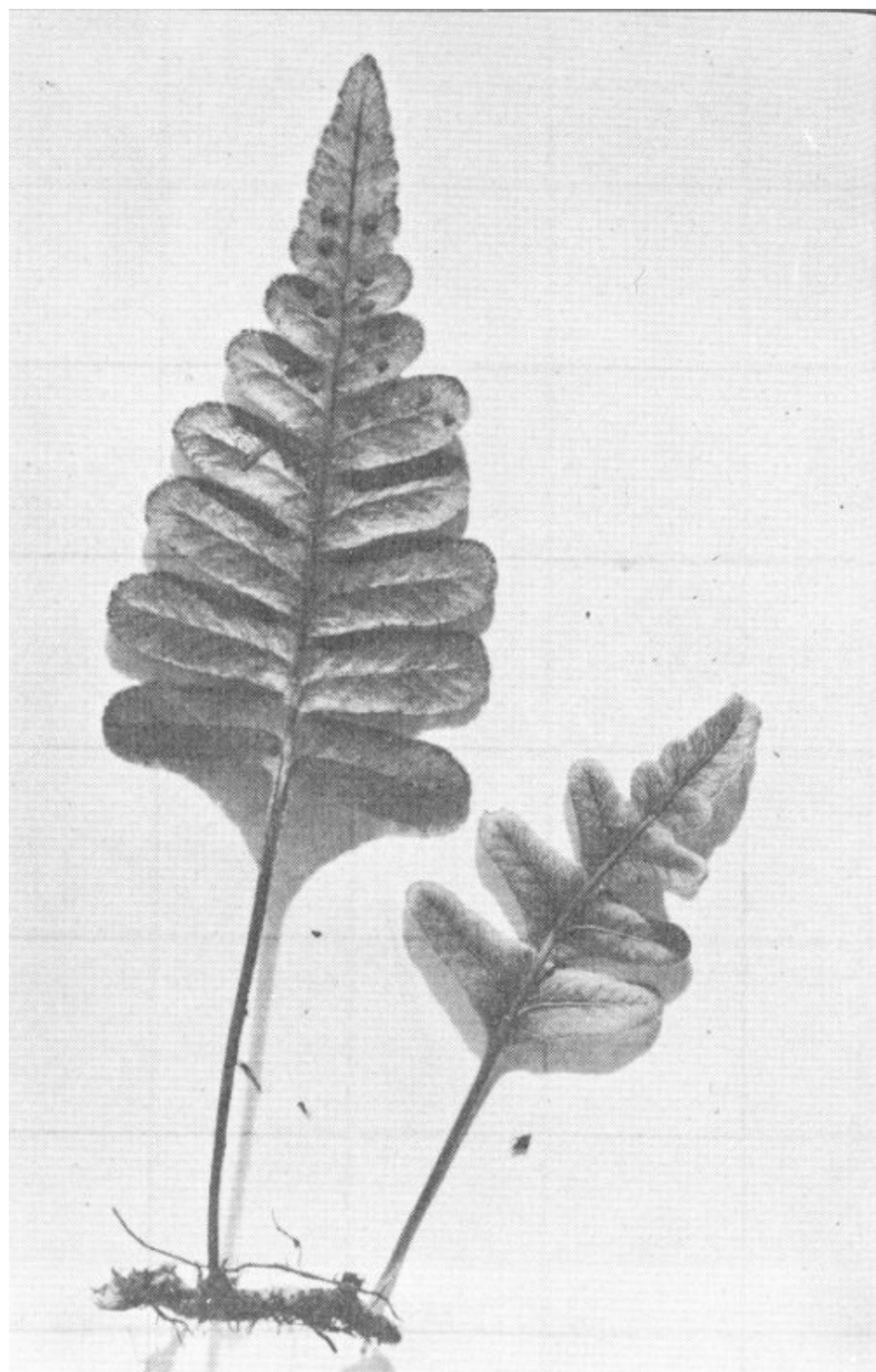
KAAPEAPE. D, (27) (pages [34](#) and [37](#))

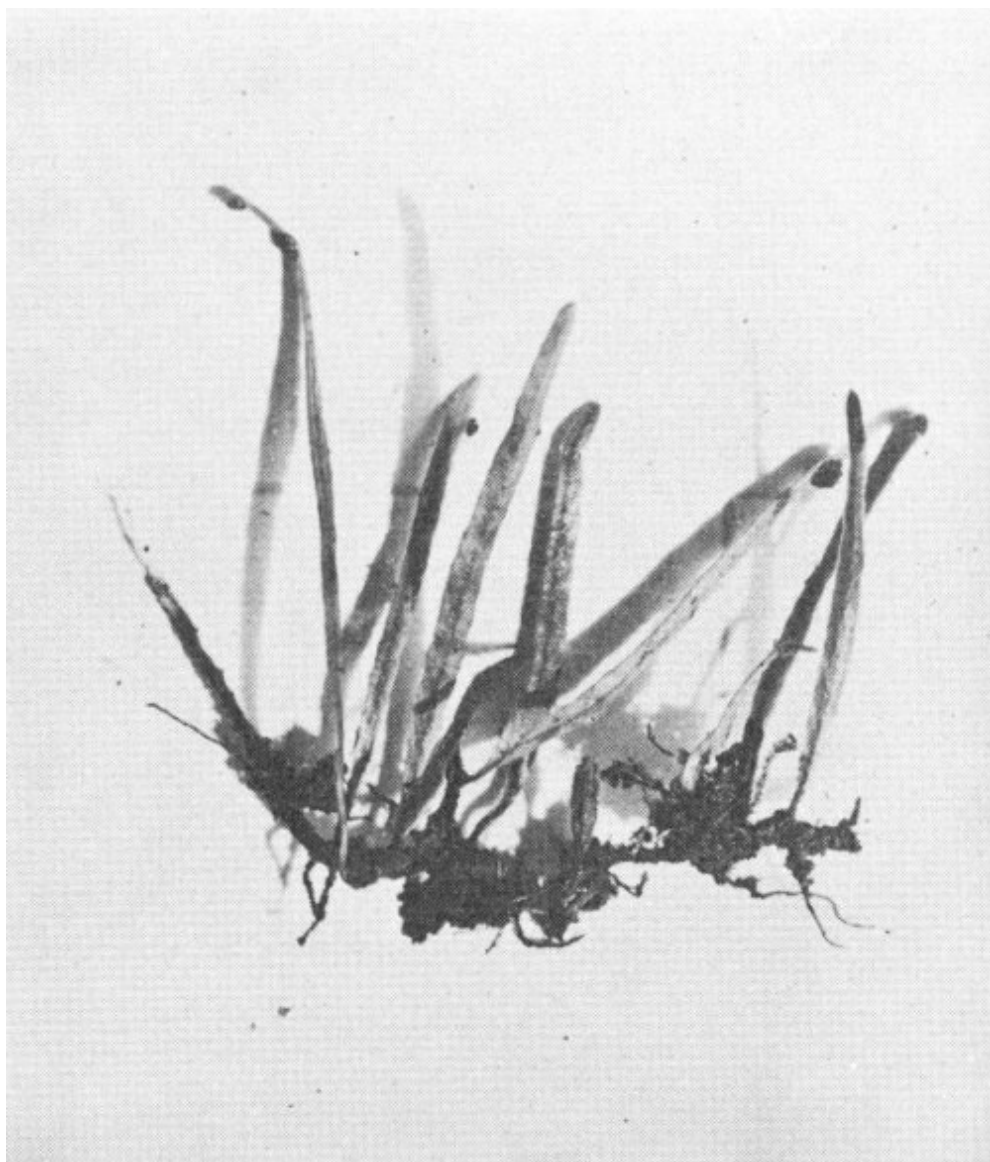


LAUKAHI. D, (32) (pages [34](#) and [38](#))

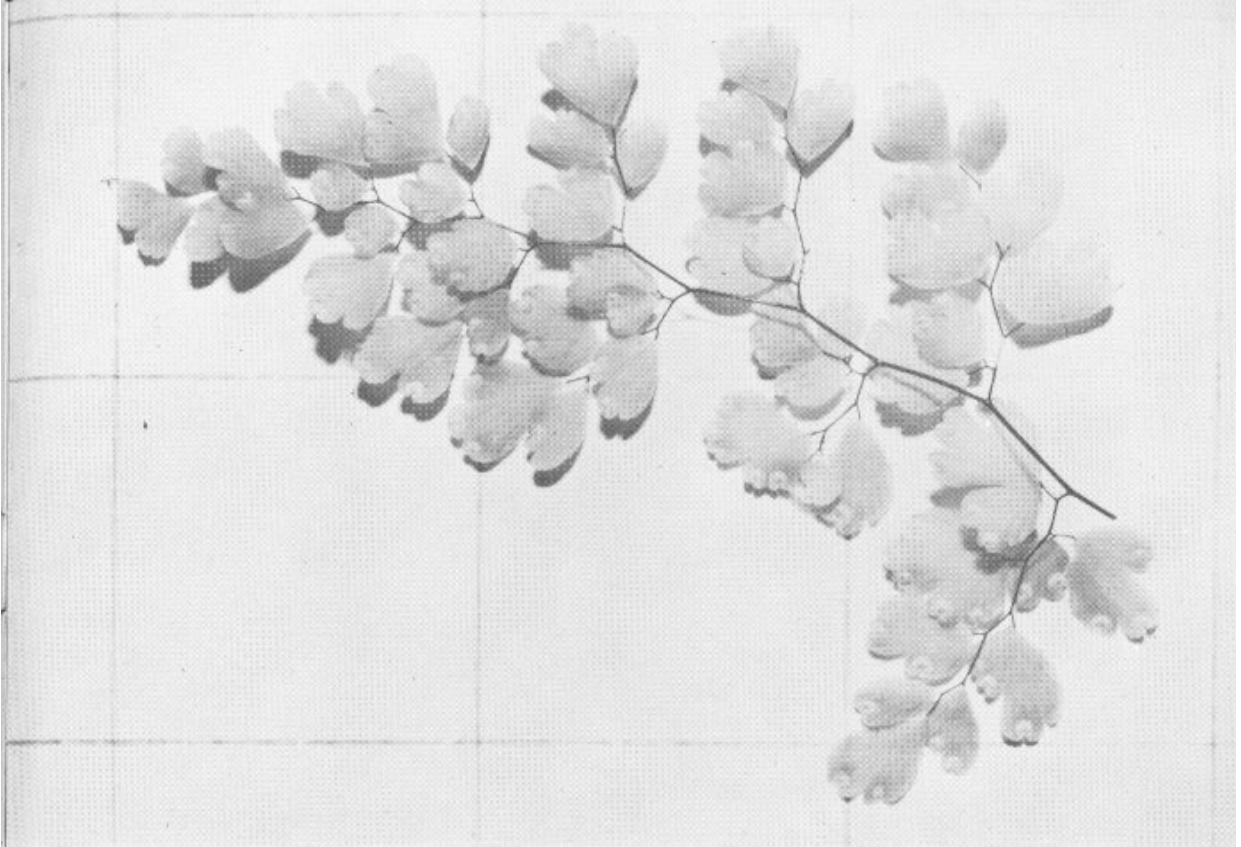


Above and below: *AE* or *pellucid polypody*. D&M, (50) (pages [36](#) and [38](#))

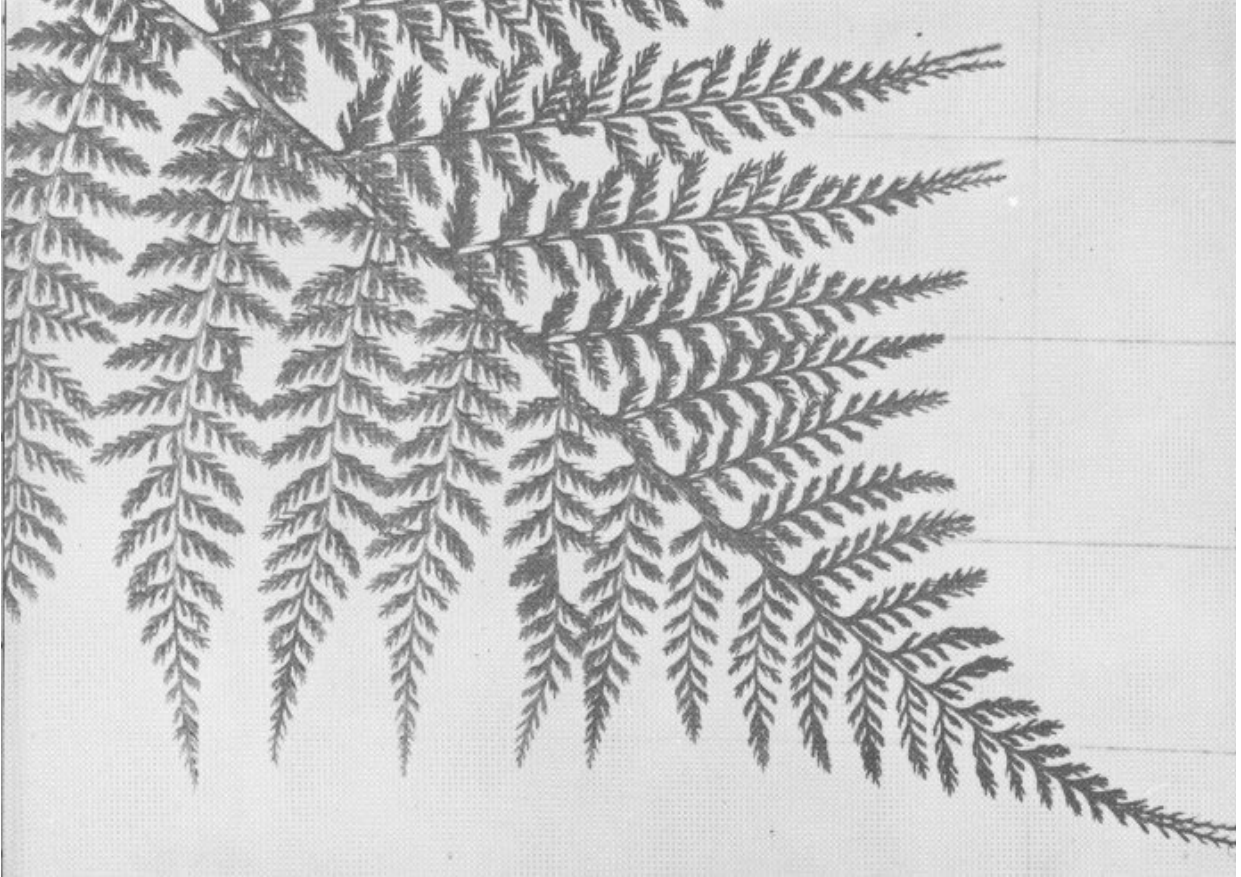




KOLOKOLO. M, (53) (pages [36](#) and [38](#))



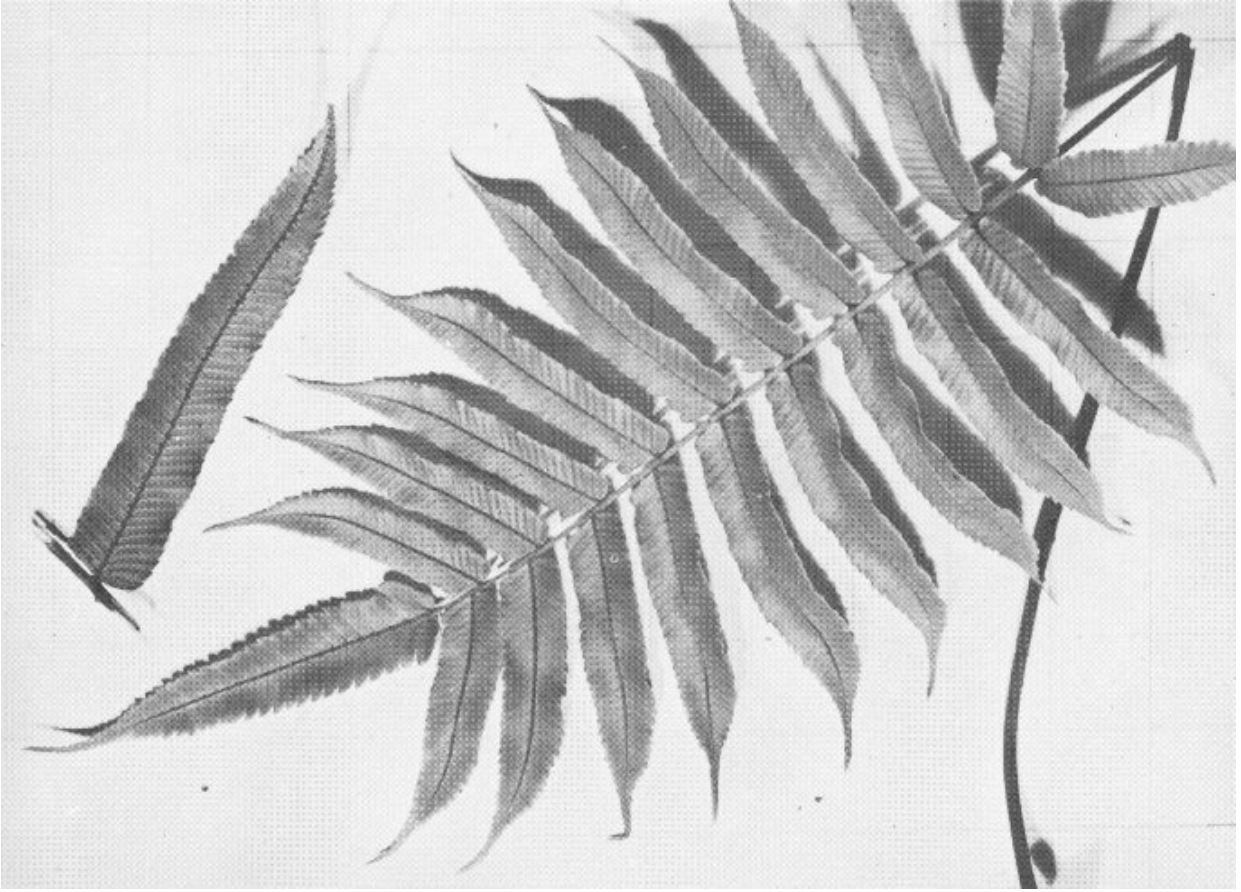
Iwaiwa or Maidenhair. M, (23) (pages [34](#) and [37](#))



AKOLEA. M, (38) (pages [35](#) and [38](#))



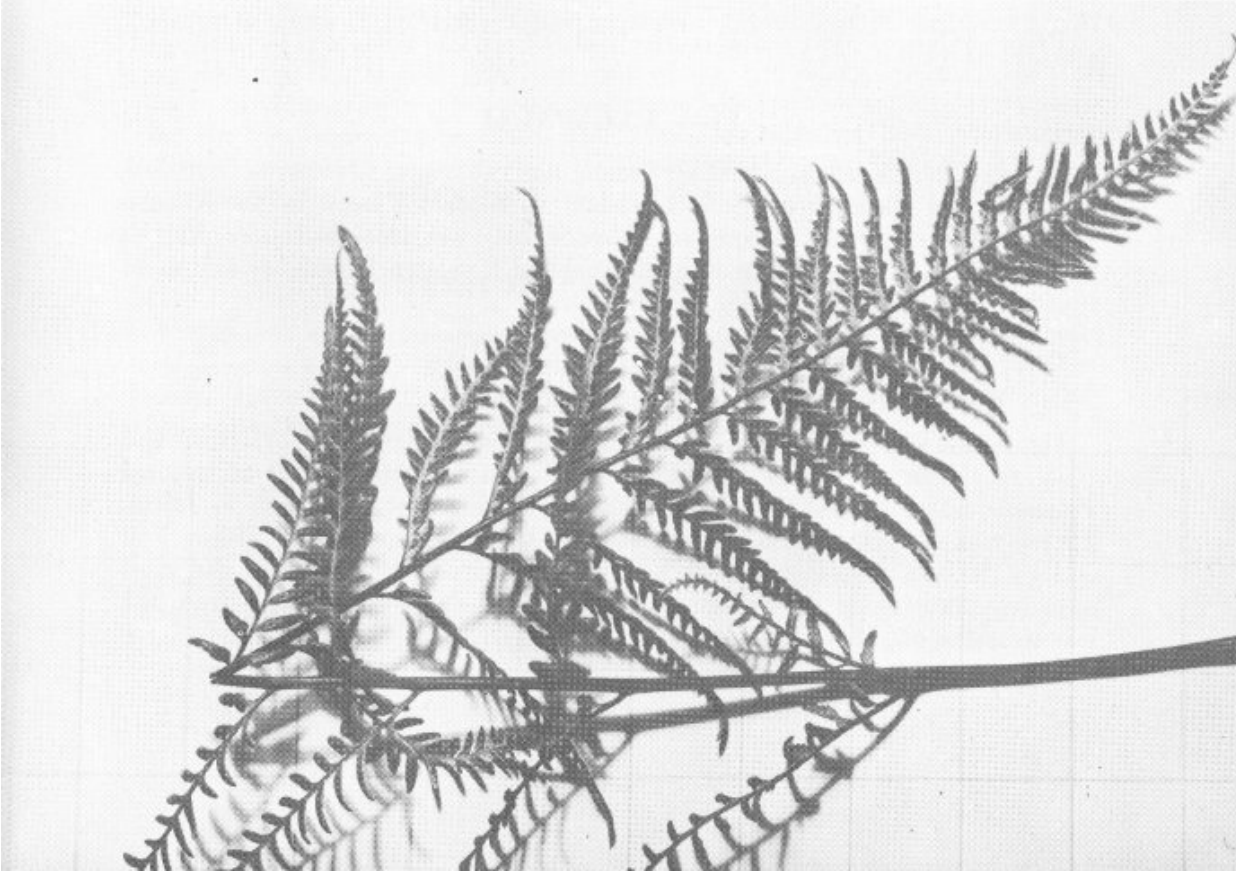
Horo. M, (39) (pages [35](#) and [38](#))



KIKAWEO. M, (35) ([page 38](#))



Cyclosorus dentatus. M, (37) (pages [35](#) and [38](#))



Silver Fern. D, (22) (pages [33](#) and [37](#))

(1-3) ADDER’S TONGUE

Sharp-eyed observers may find in cracks near Halemaumau and along the Chain of Craters Road the adder’s tongue, a small, delicate plant with a “head” which resembles a pointed tongue. Fowler (4) found the LAUKAHI growing on trees near Keanakakoi and Makaopuhi craters. Hawaiians are said to have prepared a cough remedy from the LAUKAHI. Near the 6,000 foot level on the slopes of Haleakala may be found a species of *Ophioglossum* according to Hillebrand (5).

(4) ULUHE OR FALSE STAGHORN

An attractive native fern which often becomes a pest by choking out other plants is the ULUHE, common in openings throughout the eastern half of the park. This fern often forms thickets so dense as to be virtually impenetrable. Seeds of native plants which germinate beneath these thickets may not be able to penetrate to sunlight. A fire hazard is also presented by the dried fronds. An infusion was prepared from the ULUHE, which the Hawaiians drank as an emetic, according to Fowler (4, p. 10). Hillebrand (5, p. 545) states that this fern is "common on all islands from 600 feet above the sea (Hilo district) to 3,000 feet.... The species occurs in most tropical countries of both hemispheres and in many islands of the Pacific."

(5-8) THE FILMY FERNS

Sharp eyes will be needed to detect the filmy ferns, since they normally are found on tree trunks and rocks in the jungles, and may easily be confused with the mosses with which they commonly grow. Compare the size of the dainty PALAILAULII, which is about one inch long, with the tree ferns, which often reach a height of 40 feet.

(13) PALAPALAI

An attractive fern with lace-like fronds, the PALAPALAI prefers the outskirts of woods and open places. It is abundant in Kipuka Puauulu in the Kilauea Section. The PALAPALAI is common in moderately wet areas and, according to Hillebrand (5, p. 625) is found on all inhabited Hawaiian Islands, as well as in India, Ceylon, Malaysia, Japan and Formosa.

(14) PALAA

The PALAA was called the most common of all Hawaiian ferns by Hillebrand (5, p. 627). In addition to Hawaii, he states that it is spread over all tropical Polynesia and Asia, extending as far east as Madagascar and north to Japan. Thriving in dry areas as well as moist, the PALAA is common near Hilina Pali

and in the vicinity of steam cracks around Kilauea. It is one of the most abundant ferns along the Steaming Bluffs. A red dye was extracted from the leaves by the old Hawaiians.

(15) KILAU OR BRACKEN

One of the most common ferns in dry areas in the Park is the KILAU, or bracken fern. It often may be seen growing among tall grass, and is abundant in the vicinity of Kipuka Puaulu and Kipuka Nene. It extends to about the 9,000-foot level on the slopes of Mauna Loa.

(16) OWALII OR CRETAN BRAKE

Fairly common in the Kilauea Section, this fern may be found in Kipuka Puaulu, along the Sandalwood Trail, and in the vicinity of the Thurston Lava Tube. Note the dark rows of spores along the outer edges of the leaves.

(17) WAIMAKANUI

Although not listed as a common fern, the WAIMAKANUI may be seen in open areas in Kipuka Puaulu, as may the other ferns on the same page. It is the same genus as the OWALII, described above.

(19) LOULU

This attractive fern is common near the entrance to the Thurston Lava Tube. Found in moist localities at 3,000 to 5,000 feet elevation in the Kilauea section, it normally grows as a short-stemmed plant, but may reach heights of more than three feet. Hillebrand (5, p. 550) states that it occurs also on Maui and Kauai at these elevations.

(20) LAUKAHI OR CLIFFBRAKE

Found in both sections of Hawaii National Park, the laukahi is found most frequently in dry areas. In the Kilauea section it has been found at elevations ranging up to 9,500 feet. The segments of the fronds, known technically as PINNAE, are blue-gray in color and are often shaped like a clover leaf, particularly when the fern is growing in shade. The PINNAE are usually rolled up lengthwise, hiding the reproductive structures. Hillebrand (5, p. 633) lists this species as occurring on Maui, Hawaii and Kauai, and over the High Andes from Chile to Mexico.

(21) DORYOPTERIS OR LANCE FERN

Native only to the Hawaiian Islands but having no Hawaiian name, the lance fern is common from Hilina Pali to the seacoast and throughout the Kau Desert. The plants are small, tufted, and from six to fifteen inches high. They often are abundant around the bases of lava outcroppings.

(22) SILVERFERN

A non-native plant which escaped from garden cultivation, the silverfern often grows in large clumps in dry washes in the Hilina Pali area, and extends out onto the Kau Desert. Also called “goldfern” these plants derive their name from a waxy gold or silver-colored powder on the underside of the fronds.

(23) IWAIWA OR MAIDENHAIR

Maidenhair fern, often used in floral decoration, is a plant familiar to many. Preferring shady, moist places it is common in the vicinity of Hilina Pali. The black glossy stalks provided ornamentation for Hawaiian baskets and hats. The native species apparently is being replaced by a non-native which was introduced about 50 years ago.

(24-26) NIANIAU OR SWORD FERN

The sword fern is a familiar plant in tropical and sub-tropical areas. A common house fern, its scientific name, *Nephrolepis*, means “kidney-scale” in Greek. The fern was so named because the spore dots on the underside of the leaflets are protected by a kidney-shaped tissue called an *indusium*. In the Haleakala section, *Puu Nianiau*, the hill below the lodge, received its name from these ferns, which are also called “OKUPUKUPU.” They are common in humid regions of the park, particularly in the vicinity of steam cracks near Kilauea. The three species illustrated have been found in the Park. Several large stands may be seen in Kipuka Puaulu (Bird Park).

(27) KAAPEAPE

The KAAPEAPE may be recognized by the three pointed terminal leaflet and remaining leaflets which are shaped like spear points. A species found in India and China as well as the Hawaiian Islands, the kaapeape is fairly common in Kipuka Puaulu. Note the irregular spore dots, often abundant on the undersides of the leaves.

(28-31) EKAHA OR MAUI’S PADDLE

Common throughout forested areas in the park are the plants known scientifically as *Elaphoglossum*, (Greek for “elephant-tongue”). They are usually found growing as *epiphytes* or air plants on the trunks of trees. The resemblance to a paddle was recognized by the Hawaiians as the name HOEA MAUI or “Maui’s paddle” indicates. In contrast to this smooth, shiny species is the fuzzy EKAHA. According to Hillebrand, (5, p. 549) these plants are found on all the inhabited Hawaiian Islands and in tropical America, Tahiti, India, tropical Africa, Madeira and the Azores. Six species of *Elaphoglossum* have been found in the Kilauea section.

(32) LAUKAHI

The LAUKAHI is a fairly common, low fern in the vicinity of Kipuka Puaulu, but is considered rare in other portions of the Park. Fagerlund (3, p. 19) lists it as having been collected in a lava tube at 8,800 feet on Mauna Loa.

(34) PALAPALAI O KAUMAAPUA

One of the most interesting things about this fern is the fact that the frond tapers in both directions—toward the tip and toward the base. This may be seen in the illustration. This species is fairly abundant in the crater near the Thurston Lava Tube.

35

(36) CYCLOSORUS SANDWICENSIS

Although this fern is a native Hawaiian species, no Hawaiian name for it could be found. It prefers shade and moist areas and is fairly common in the Thurston Lava Tube and Kilauea Iki trails.

(37) CYCLOSORUS DENTATUS

The fern known technically as *Cyclosorus dentatus* is not native to Hawaii, but apparently was considered common on Oahu after 1900, according to Wagner (6, p. 110). It may be found in moist areas in and near the Park.

(38) AKOLEA

One of the most dainty of ferns, the AKOLEA is found in wet areas in the vicinity of the Thurston Lava Tube. It is a common fern on the Kulani Trail, just outside of the Park.

(39) HOIO

Common in moist areas in the Kilauea section, the HOIO prefers shade. The fronds are large, often 3 to 4 feet long. Below the branches the midribs are dark brown and smooth, but clothed with dark scales at their bases. Spore dots are abundant.

(42) EKAHA OR BIRDNEST FERN

Of all Hawaiian fern species, few are more beautiful than the birdnest fern, whose fronds often reach a length of six feet. Although it has not been described from the Park, it has been included because it is abundant in the Puna District, and probably will be found in the Kalapana Extension of the Park. It normally grows in trees, but it may also attach itself to rocks. When this happens the young birdnest ferns are often called "rock ferns." The dark areas on the underside of the leaves in the illustration are spores.

(43-46, 49) ASPLENIUM

Among the most attractive, the *Asplenium* ferns are also among the most diversified Hawaiian plants as far as appearance is concerned. Showing wide variety in choice of habitat they range from moist jungles to arid lava flows. A number of the species may be recognized by the similarity in shape of the leaf segments, and the long, parallel rows of spores, but all do not show these characteristics. Nine species have been found in the Kilauea Section of the Park.

(47) OWALII OR MAIDENHAIR SPLEENWORT

The maidenhair spleenwort has adapted itself to a dry existence. It is abundant on barren lava flows where few other plants grow, and ranges up to an elevation of about 10,000 feet.

(48) PAMOHO

A fern which lives in moist shady craters and lava tubes, the PAMOHO may be seen growing on the cliff at the mouth of the Thurston Lava Tube.

36

(50) AE OR PELLUCID POLYPODY

A common plant throughout the open forests of the Kilauea Section, this fern may be seen growing in abundance at the bases of ohia trees. Superficially resembling a sword fern, the sharp twists to the leaf segments differentiate it from them. An interesting thing about this species is that it has also adapted itself to a life as an epiphyte, or air plant, in the rain forests, at which time it develops a heavy root-stock and has a much different appearance, as may be seen in the illustrations.

(51-57) KIHĪ AND RELATIVES

The six ferns illustrated were formerly grouped under the genus *Polypodium*, but several have recently been reclassified and placed in new genera. The plants illustrated normally grow as *epiphytes*, or air plants on tree trunks and rocks in the rain forests of the Kilauea Section. That the Hawaiians recognized them as distinct plants is indicated in the fact that they were given different Hawaiian names.

(58-61) WAWAEIOLE OR CLUBMOSS

Clubmosses are sometimes called “living fossils.” Many deposits of the continental United States are formed from giant relatives of these plants. The name “clubmoss” is derived from the spore-bearing bodies, which are club-shaped. Most abundant is the nodding clubmoss but at least three other species may be found in the Kilauea and Haleakala sections.

Degener (2) reports that the Hawaiians used the wawaeiole (“rat’s foot”) as a relief for rheumatism, the sufferer bathing in water in which the plants had been boiled for about three hours.

(62-63) MOA OR PIPI

The *Psilotum* is a primitive plant, of which two species are found in the Hawaiian Islands. One is the MOA, which usually grows on the ground and has stems which fork many times near the ends. The second, the PIPI, often grows upon the trunks of trees and ferns. This species normally grows perpendicular to the trunk, then curves abruptly earthward like the tail of a horse. Degener (2, p. 18) states that the MOA was used by the Hawaiians in making a medicine for treatment of a fungus disease, as an emetic, and that the spores were used as a powder to relieve chafing from the malo, or loincloth.

SCIENTIFIC NAMES OF FERNS DESCRIBED HEREIN

(Numbers follow common names in the captions. Because of numerous recent studies of ferns, various name changes have been necessary, and on some names there is still disagreement. The following list adopts mainly the nomenclature of the GENERA FILICUM by Prof. Edwin B. Copeland, 1947. The indented names are ones which have been used for the same plants in the past.)

FAMILY OPHIOGLOSSACEAE

- (1) *Ophioglossum petiolatum* Hooker
- (2) *Ophioglossum nudicaule* Linnaeus fils
- (3) *Ophioglossum pendulum* esp. *falcatum* (Presl) Clausen (LAUKAHI)
 - O. falcatum* (Presl) Fowler
 - O. pendulum* L.
 - Ophioderma falcatum* Degener

FAMILY GLEICHENIACEAE

- (4) *Dicranopteris emarginata* (Brackenridge) Robinson (ULUHE)
 - D. sandwicensis* Degener
 - Gleichenia emarginata* (Brackenridge) Robinson

FAMILY HYMENOPHYLLACEAE

- (5) *Hymenophyllum obtusum* Hooker and Arnott (PALAILAULII)
 - Sphaerocionium obtusum* (H.&A.) Copeland
- (6) *Hymenophyllum lanceolatum* Hooker and Arnott (PALAIHINAHINA)
 - Sphaerocionium lanceolatum* (H.&A.) Copeland

- (7) *Hymenophyllum recurvum* Gaudichaud (OHIAKU)
Mecodium recurvum (Gaud.) Copeland
- (8) *Trichomanes davallioides* Gaudichaud (KILAU)
Vandenboschia davallioides (Gaud.) Copeland

FAMILY PTERIDACEAE

- (9) *Cibotium Chamissoi* Kaulfuss (HAPU III)
C. Menziesii Hooker
- (10) *Cibotium splendens* (Gaudichaud) Krajina (HAPUU)
C. Chamissoi Kaulfuss
- (11) *Cibotium glaucum* (Smith) Hooker and Arnott (HAPUU)
- (12) *Cibotium hawaiiense* Nakai and Ogura (MEU)
Cibotium St. Johnii Krajina
- (13) *Microlepia setosa* (Smith) Alston (PALAPALAI)
M. strigosa (Thunberg) Presl
- (14) *Sphenomeris chusana* (Linnaeus) Copeland (PALAA)
Stenoloma chinensis (L.) Beddome
Odontosoria chinensis (L.) J. Smith
Microlepia tenuifolia Mettenius
- (15) *Pteridium aquilinum* (L.) Kuhn (KILAU, BRACKEN)
Pteris aquilina L.
- (16) *Pteris cretica* Linnaeus (OWALII)
- (17) *Pteris excelsa* Gaudichaud (WAIMAKANUI)
- (18) *Pteris vittata* Linnaeus (KILAUOPUEO)
Pteris longifolia L.
- (19) *Coniogramme pilosa* (Brackenridge) Hieronymus (LOULU)
Gymnogramme javanica Hillebrand
- (20) *Pellaea ternifolia* (Cavanilles) Link (LAUKAHI)
- (21) *Doryopteris decora* Brackenridge (LANCE FERN)
- (22) *Pityrogramma calomelanos* (Linnaeus) Link (SILVER FERN)
Ceropteris ochracea (Presl) Robinson
- (23) *Adiantum cuneatum* Langsdorff and Fischer (IWAIWA, MAIDENHAIR)

FAMILY DAVALLIACEAE

- (24) *Nephrolepis exaltata* (Linnaeus) Schott (NIANIAU, COMMON SWORDFERN)
(25) *Nephrolepis cordifolia* (Linnaeus) Presl (NARROW SWORDFERN)
(26) *Nephrolepis hirsutula* (Forster) Presl (SCALY SWORDFERN)

FAMILY ASPIDIACEAE

- (27) *Cyrtomium caryotideum* (Wallich) Presl (KAAPEAPE)
 Aspidium caryotideum Wallich
 Phanerophlebia caryotideum (Wall.) Copeland
(28) *Elaphoglossum aemulum* (Kaulfuss) Brackenridge (EKAHA)
 E. conforme (Swartz) Schott
 Acrostichum conforme Swartz
(29) *Elaphoglossum gorgoneum* (Kaulfuss) Brackenridge (EKAHA)
(30) *Elaphoglossum hirtum* (Swartz) C. Christensen (EKAHA)
 Acrostichum squamosum Hillebrand
(31) *Elaphoglossum reticulatum* (Kaulfuss) Gaudichaud (EKAHA)
(32) *Dryopteris paleacea* (Swartz) C. Christensen (LAUKAHI)
 Aspidium filix-mas var. *parallelogrammum* Kuntze
(33) *Dryopteris glabra* (Brackenridge) O. Kuntze (KILAU)
 Aspidium glabrum (Brack.) Mettenius
(34) *Lastrea globulifera* Brackenridge (PALAPALAI O KAUMAAPUA)
 Aspidium globuliferum (Brack.) Mann
(35) *Cyclosorus cyatheoides* (Kaulfuss) Farwell (KIKAWEO)
 Dryopteris cyatheoides (Kaulf.) O. Kuntze
 Aspidium cyatheoides Kaulf.
(36) *Cyclosorus sandwicensis* (Brackenridge) Copeland
 Dryopteris stegnogrammoides (Baker) C. Christensen
 Phegopteris polycarpa (Hooker & Arnott) C. Chr.
(37) *Cyclosorus dentatus* (Forsk.) Ching
 Dryopteris dentata (Forsk.) O. Kuntze
(38) *Athyrium microphyllum* (Smith) Alston (AKOLEA)
 A. Poiretianum (Gaudichaud) Presl
 Asplenium aspidioides Hillebrand

- A. multisectum* Brackenridge
(39) *Athyrium sandwichianum* Presl (HOIO)
Diplazium sandwichianum (Presl) Diels
Asplenium sandwichianum (Presl) Hooker

FAMILY BLENNACEAE

- (40) *Sadleria cyatheoides* Kaulfuss (AMAUMAU)
(41) *Sadleria pallida* Hooker & Arnott (AMAU)
S. Hillebrandii Robinson

FAMILY ASPLENIACEAE

- (42) *Asplenium nidus* Linnaeus (EKAHA, BIRDNEST FERN)
Neottopteris nidus (L.) J. Smith
Thamnopteris nidus Presl
(43) *Asplenium contiguum* Kaulfuss
(44) *Asplenium rhipidoneuron* Robinson (IWAIWA O KANE)
A. furcatum Hillebrand
(45) *Asplenium lobulatum* Mettenius (PIPIILAU MANAMANA)
A. pseudofalcatum Hillebrand
(46) *Asplenium Macraei* Hooker and Greville
A. erectum Hillebrand
(47) *Asplenium trichomanes* Linnaeus (OWALII)
A. densum Brackenridge
(48) *Asplenium unilaterale* Lamarck (PAMOHO)
A. resectum Smith
(49) *Asplenium adiantum-nigrum* Linnaeus (IWAIWA)

FAMILY POLYPODIACEAE

- (50) *Polypodium pellucidum* Kaulfuss (AE)
(The variety of exposed places is var. *vulcanicum* Skotts.)

- (51) *Pleopeltis Thunbergiana* Kaulfuss (EKAHA AKOLEA)
Polypodium lineare Thunberg

FAMILY GRAMMITIDACEAE

- (52) *Grammitis Hookeri* (Brackenridge) Copeland
Polypodium Hookeri Brackenridge
- (53) *Grammitis tenella* Kaulfuss (KOLOKOLO)
Polypodium pseudogrammitis Gaudichaud
- (54) *Xiphopteris Saffordii* (Maxon) Copeland (KIH)
Polypodium Saffordii Maxon
P. serrulatum Hillebrand
- (55) *Amphoradenium hymenophylloides* (Kaulfuss) Copeland (PAI)
Polypodium hymenophylloides Kaulfuss
- (56) *Amphoradenium sarmentosum* (Brackenridge) Copeland
Polypodium sarmentosum Brackenridge
- (57) *Amphoradenium tamariscinum* (Kaulfuss) Copeland (WAHINI NOHO
MAUNA)
Polypodium tamariscinum Kaulfuss

FAMILY LYCOPODIACEAE

- (58) *Lycopodium cernuum* Linnaeus (WAWAEIOLE)
- (59) *Lycopodium phyllanthum* Hooker & Arnott (WAWAEIOLE)
Lycopodium pachystachyon Spring
- (60) *Lycopodium polytrichoides* Kaulfuss (WAWAEIOLE)
- (61) *Lycopodium venustulum* Gaudichaud (WAWAEIOLE)

FAMILY PSILOTACEAE

- (62) *Psilotum complanatum* Swartz (MOA)
- (63) *Psilotum nudum* (Linnaeus) Beauvois (PIPI)
Psilotum triquetrum Swartz

SELECTED BIBLIOGRAPHY

- (1) Copeland, Edwin B. *Genera Filicum*. Waltham, Mass., 1947.
- (2) Degener, Otto. *Plants of Hawaii National Park*. Honolulu, 1930.
- (3) Fagerlund, Gunnar, and Arthur L. Mitchell. *A Checklist of the Plants Hawaii National Park, Kilauea-Mauna Loa Section*. Hawaii National Park, 1944.
- (4) Fowler, Robert. "Annotated List of Ferns of the Kilauea-Mauna Loa Section of Hawaii National Park." *American Fern Journal*, Vol. 30, No. 1, 1940.
- (5) Hillebrand, W. F. *Flora of the Hawaiian Islands*. New York, 1888.
- (6) Wagner, W. H., Jr. "Ferns Naturalized in Hawaii." *Occasional Papers of Bernice P. Bishop Museum*, Vol. XX, No. 8. Honolulu, 1950.

INDEX

[A](#) [B](#) [C](#) [D](#) [E](#) [F](#) [G](#) [H](#) [I](#) [J](#) [K](#) [L](#) [M](#) [N](#) [O](#) [P](#) [Q](#) [R](#) [S](#) [T](#) [U](#) [V](#) [W](#)
[X](#) [Y](#) [Z](#)

A

Adder's tongue [17](#), [32](#), [37](#)
Adiantum [29](#), [34](#), [37](#)
Ae [28](#), [36](#), [38](#)
Akolea [29](#), [35](#), [38](#)
Amau [7](#)
Amaumau [7](#), [20](#)-21
Amphoradenium [18](#), [36](#), [38](#)
Asplenium [10](#), [13](#), [24](#), [35](#), [38](#)
Athyrium [29](#), [35](#), [38](#)

B

Bird nest fern [22](#), [35](#), [38](#)
Bracken [13](#), [33](#), [37](#)

C

Ceropteris [31](#), [33](#), [37](#)
Cibotium [4](#), [5](#), [37](#)
Cliffbrake [12](#), [33](#), [38](#)
Clubmoss [8](#), [36](#), [39](#)
Coniogramme [11](#), [33](#), [37](#)
Cretan brake [26](#), [33](#), [37](#)
Cyclosorus [22](#), [31](#), [35](#), [38](#)
Cyrtomium [27](#), [34](#), [37](#)

D

Dicranopteris [9](#), [32](#), [37](#)
Diplazium [30](#), [35](#), [38](#)
Doryopteris [12](#), [33](#), [37](#)
Dryopteris [27](#), [34](#), [38](#)

E

Ekaha	22 , 23 , 34 , 38
Ekaha akolea	18 , 36 , 38
<i>Elaphoglossum</i>	23 , 34 , 38
Epiphytes	18

F

False staghorn	9 , 32 , 37
----------------	---

G

<i>Gleichenia</i>	9 , 32 , 37
Gold fern	31 , 33 , 37
<i>Grammitis</i>	18 , 36 , 38

H

Hapu iii	4 , 5 , 37
Hapuu	4 , 5 , 37
Haleakala	3 , 15
Hawaii National Park	3
Hoi	30 , 35 , 38
<i>Hymenophyllum</i>	18 , 32 , 37

I

Indusium	34
Iwaiwa	24 , 35 , 38
Iwaiwa O Kane	25 , 35 , 38

K

Kaapeape	27 , 34 , 37
Kapa cloth	7
Keanakakoi Crater	7
Kihi	18 , 36 , 38
Kikaweo	30 , 38
Kilau	13 , 18 , 32 , 33 , 37
Kilauea	3
Kipuka Puauulu	33 , 34
Koa	3
Kolokolo	28 , 36 , 38

L

Lance fern	12 , 33 , 37
Laukahi	12 , 27 , 33 , 34 , 37 , 38
Loulu	11 , 33 , 38
<i>Lycopodium</i>	8 , 36 , 39

M

Maidenhair	29 , 34 , 37
Maidenhair spleenwort	13 , 35 , 38
Makaopuhi Crater	6 , 32
Maui's paddle	23 , 34 , 38
Mauna Loa	3
Meu	4 , 5 , 37
<i>Microlepia</i>	23 , 32 , 37
Moa	14 , 36 , 39
Mother of ohia	4

N

Napau Crate	6
National Park Service	3
<i>Neottopteris</i>	22 , 35 , 38
<i>Nephrolepis</i>	16 , 34 , 37
<i>Niania</i>	16 , 34 , 37

O

<i>Odontosoria</i>	14 , 32 , 37
Ohiaku	18 , 32 , 37
<i>Ophioderma</i>	17 , 32 , 37
<i>Ophioglossum</i>	17 , 32 , 37
Owalii	13 , 26 , 33 , 35 , 37 , 38

P

Pai	18 , 36 , 38
Palaa	9 , 14 , 32 , 37
Palaihinahina	18 , 32 , 37
Palailaulii	18 , 32 , 37
Palapalai	11 , 23 , 32 , 34 , 37 , 38
Pamoho	10 , 35 , 38

<i>Pellaea</i>	12 , 33 , 38
Pellucid polpody	28 , 36 , 38
Piipiilau manamana	25 , 35 , 38
Pinnae	33
Pipi	14 , 36 , 39
<i>Pityrogramma</i>	31 , 33 , 37
<i>Polypodium</i>	18 , 28 , 36 , 38
<i>Psilotum</i>	14 , 36 , 39
<i>Pteridium</i>	13 , 33 , 37
<i>Pteris</i>	26 , 33 , 37
Pulu	4 , 6
Pulu Factory	6 , 8

S

<i>Sadleria</i>	7 , 20 - 21
Silverfern	31 , 33 , 37
<i>Spenomeris</i>	14 , 32 , 37
Spores	3
Staghorn, false	9 , 32 , 37
<i>Stenoloma</i>	14 , 32 , 37
Sword fern	16 , 34 , 37

T

Thurston Lava Tube	10 , 33 , 34
Tree fern	2 , 4 , 5 , 37
<i>Trichomanes</i>	18 , 32 , 37

U

Uluhe	9 , 32 , 37
-------	---

W

Wahini noho mauna	18 , 36 , 38
Waimakanui	26 , 33 , 37
Wawaeiole	8 , 36 , 39

X

<i>Xiphopteris</i>	18 , 36 , 38
--------------------	--

**THE PURPOSES OF THE
HAWAII NATURAL HISTORY ASSOCIATION
Hawaii National Park, Hawaii**

Formed for the purpose of cooperating with the National Park Service by assisting the Naturalist Department of Hawaii National Park in the development of a better public understanding of the volcanology, geology, plant life, history, and related interests in Hawaii National Park and surrounding areas, the Association aids in the development of the Hawaii National Park Museum and library, offers publications, motion and still colored pictures of volcanoes and natural history of this area for sale to park visitors, and cooperates in the publication of Hawaii Nature Notes.

Revenue derived from the activities of the Hawaii Natural History Association is devoted entirely to the purposes outlined above. Any person interested in furthering these purposes may become a member upon payment of one dollar. Gifts and donations are accepted for museum or library development, or general use.

Transcriber's Notes

- Silently corrected a few typos.
- Added captions (derived from the text, and enclosed in {brackets}) to several illustrations.
- Retained publication information from the printed edition: this eBook is public-domain in the country of publication.
- In the text versions only, text in italics is delimited by _underscores_.

*** END OF THE PROJECT GUTENBERG EBOOK FERNS OF
HAWAII NATIONAL PARK ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.